

NK cells for cancer: phenotypes, engineering and manufacturing that improve efficacy

A literature synthesis across research and process-development evidence

313 curated records, 1993–2026 · 50 clinical trials · 58 process/manufacturing
methods · 138 preclinical/mechanistic · 39 reviews · 244 with culture/
manufacturing content

Overview — NK cells for cancer: phenotype, engineering and manufacturing

Scope

This collection curates **313 papers (1993-2026; 94 from 2024-2026, 166 from 2020 onward)** on what actually makes an NK-cell cancer therapy work: which **cell state** you infuse, which **genetic modifications** you install, and which **culture and manufacturing process** you use to get there. Manufacturing is treated as primary evidence rather than a methods footnote — 58 papers are classified as process/method studies and 244 carry culture or manufacturing content.

Evidence class	n
Preclinical (in vivo)	105
Process / manufacturing methods	58
Clinical trials	50
Reviews	39
Mechanistic / in vitro	33
Human translational	28

Open-access full-text XML was retrieved for 153 records (prioritising trials, process papers and human studies); the remainder are catalogued from title, abstract and identifiers.

Topic axes

Axis	n	Core question
car_nk_engineering	32	Receptor architecture, costimulation, gene delivery into NK cells
expansion_feeders	30	Feeders, media, vessels, bioreactors, GMP scale-up
clinical_nk	28	What has actually been observed in patients
nk_phenotype_states	26	Which NK state to select, expand or induce
gene_editing_knockouts	24	Which brakes to delete (CISH, NKG2A, TGF- β /SMAD, CD38, Regnase-1...)
cytokine_support	24	IL-15/IL-21/IL-12+15+18 priming and cytokine armouring
general_nk_therapy	22	Cross-cutting landscape and NK biology fundamentals
nk_cell_sources	20	Peripheral blood, cord blood, iPSC, NK-92
nk_engagers	20	BiKE/TriKE/NKCE molecules that supply specificity without gene transfer
metabolism_persistence	18	Metabolic fitness, hypoxia, survival after infusion
checkpoints_inhibitory	18	NKG2A/HLA-E, TIGIT, PD-1, KIR, TIM-3
adcc_combinations	18	CD16 biology, antibody and drug combinations
trafficking_solid_tme	18	Homing, infiltration, suppression inside solid tumours

Axis	n	Core question
cryo_qc_manufacturing	15	Cryopreservation damage, potency assays, release testing

Disease context is dominated by platform/pan-cancer work (149), then AML/MDS (25), lymphoma/CLL (24), gastrointestinal (16), myeloma (15), head-and-neck/melanoma (13) and glioblastoma (11).

The ten claims this literature supports

- 1. The infused cell state matters more than the donor.** Brief IL-12+IL-15+IL-18 preactivation converts resting NK cells into cytokine-induced memory-like (CIML) cells whose enhanced responsiveness persists for weeks (Romee 2012, PMID 22983442) and translates into remissions in AML (Romee 2016, PMID 27655849).
- 2. CD56bright/CD56dim is an inadequate description of the product.** scRNA-seq/CITE-seq resolves multiple human NK states (Rebuffet 2024, PMID 38956378) and maps tumour-infiltrating states across cancers (Netskar 2024, PMID 38956379; Tang 2023, PMID 37607536); CD56bright cells become potently cytotoxic after IL-15 priming (Wagner 2017, PMID 28972539).
- 3. The tumour actively reprogrammes the product after infusion** — TGF- β -driven conversion to ILC1-like cells (Gao 2017, PMID 28759001), hypoxia/HIF-1 α (Ni 2020, PMID 32445619), lactylation (2025, PMID 40494934), nutrient competition — so engineering increasingly targets resistance to the microenvironment rather than target recognition alone.
- 4. CAR-NK signalling is not CAR-T signalling.** NK-native transmembrane/costimulatory modules (NKG2D-TM + 2B4 + CD3 ζ ; Li 2018, PMID 30082067) and, surprisingly, non-native CD28 via LCK/CD3 ζ /ZAP70 (PMID 38900051) outperform naive CAR-T architectures ported into NK cells.
- 5. Cytokine armouring is the difference between a transient and a persistent product.** IL-15-armoured cord-blood CAR-NK cells persist and control tumours (Liu 2018, PMID 28725044) and underpin the CD19 CAR-NK trials; loss of metabolic fitness, not antigen loss, drives relapse in preclinical CAR-NK models (PMID 37494448).
- 6. Deleting brakes is now as important as adding receptors.** CISH (PMID 32531207), NKG2A/KLRC1 (PMID 39349459), SMAD4 (PMID 40119192), TGFB β 2, CD38 (PMID 32603414), Regnase-1 (PMID 38821052) and genome-wide CRISPR screens (PMID 40845844; PMID 38918616) define an actionable edit list.
- 7. Expansion platforms are converging on membrane-bound IL-21/4-1BBL feeders or their cell-free equivalents.** K562-mb15-41BBL (Imai/Fujisaki 2009, PMID 19383914), mbIL-21 aAPC and PM21 particles (PMID 27059202) give log-scale expansion; feeder-free media, G-Rex/gas-permeable vessels, CliniMACS Prodigy and stirred bioreactors carry it into GMP.
- 8. Cryopreservation is a first-order efficacy variable, not a logistics detail.** Cryopreserved NK cells lose 3D migration and cytotoxicity even when degranulation assays look normal (Mark 2020, PMID 33067467); DMSO-free formulations and post-thaw rescue strategies are an active 2024–2026 workstream.
- 9. Clinical activity is real but concentrated where the product is armoured and the target is haematological.** CD19 CAR-NK (Liu 2020, PMID 32023374; Marin

2024, PMID 38238616), iPSC-derived FT596 (PMID 39798981), 4-1BB CAR-NK (PMID 40251398), CIML NK (PMID 35349491) and AFM13-loaded cord-blood NK (PMID 40186077) show responses without CRS/ICANS or GvHD; solid-tumour benefit remains sparse.

10. **Efficacy is a product of the whole chain.** The same edit or CAR gives different results depending on feeder, medium, cytokine schedule, cryopreservation and lymphodepletion — which is why the manufacturing literature in this collection is not separable from the biology.

How to read the collection

- notes/01–10 are the narrative synthesis, ordered from cell state to clinical translation.
- REPORT.md is auto-generated from index.tsv (never hand-edited) and lists every paper by axis and evidence class.
- Papers are cited as first author, year and PMID; index.tsv carries venue, citation count, DOI, PMCID and the local full-text path where available.
- Reproduce with `python3 harvest.py && python3 curate.py && python3 fetch_fulltext.py && python3 gen_report.py && python3 build_pdf.py`.

NK phenotypes and cell states: what to infuse

The classical dichotomy is a starting point, not a specification

CD56bright (cytokine-producing, CD16-low) versus CD56dim (cytotoxic, CD16-high, KIR/CD57-acquiring) remains the working vocabulary, but the field has moved past it:

- **Freud, Mundy-Bosse & Caligiuri (2017, PMID 29166586)** argued that the human NK spectrum is far broader than the dichotomy, shaped by tissue localisation and differentiation stage.
- **Rebuffet et al. (2024, PMID 38956378)** used scRNA-seq + CITE-seq to resolve human NK heterogeneity into distinct transcriptional states with surface-protein correlates — including states not separable by the canonical marker panel.
- **Netskar et al. (2024, PMID 38956379)** built a single-cell transcriptional reference map of blood- and tissue-derived NK cells and used it to map tumour-infiltrating NK cells across cancers; **Tang et al. (2023, PMID 37607536)** provide the complementary pan-cancer single-cell panorama.
- **Horowitz et al. (2013, PMID 24154599)** established by mass cytometry that NK repertoire diversity is jointly genetically and environmentally determined — the reason donor-to-donor product variability is structural, not technical.
- **Dogra et al. (2020, PMID 32059780)** showed tissue imprinting of human NK development, function and residence, which is why blood-derived products behave differently from the tissue NK cells that a solid tumour actually contains.

Practical consequence: "CD56+CD3−, >90% pure" is not a product specification. Recent trials increasingly report high-dimensional immunophenotypic and transcriptional characterisation of the infused product (e.g. the CIML NK head-and-neck trial, PMID 39948608), and that is becoming the expected standard.

Cytokine-induced memory-like (CIML) NK cells

The most reproducible phenotype-engineering result in the field:

- **Romee et al. (2012, PMID 22983442)** — a brief (12–16 h) preactivation with IL-12 + IL-15 + IL-18 produces human NK cells with enhanced IFN-γ recall responses to cytokine or activating-receptor restimulation for weeks after the stimulus is withdrawn.
- **Romee et al. (2016, PMID 27655849)** — CIML NK cells show enhanced antileukaemic responses and, in a first-in-human trial in relapsed/refractory AML, expanded in vivo and produced remissions.
- **Berrien-Elliott et al. (2022, PMID 35349491)** — donor CIML NK cells infused for post-transplant relapse expand, persist and show efficacy, with detailed pharmacodynamics of expansion and persistence.
- **2024 mechanism (PMID 38941480)** — memory-like subsets form via **epigenetic rewiring** and transcriptional regulation, i.e. the phenotype is a stable differentiation state rather than a transient activation.

- **Combination-ready:** CIML differentiation stacks with engagers (AFM13 pre-complexing, PMID 33986022), with CARs (CAR-CIML NK against mesothelin in ovarian cancer, PMID 38996027), with IL-15 super-agonist plus CTLA-4 blockade in head-and-neck cancer (PMID 39948608), and with PM21-particle expansion (PMID 38711506), which reports memory-like characteristics plus enhanced survival from a single manufacturing process.

Adaptive / NKG2C+ NK cells

- HCMV-driven **CD56dim CD57+ NKG2C+** adaptive NK expansions are associated with reduced leukaemia relapse after reduced-intensity transplant (Cichocki 2016, PMID 26416461); the drivers of that expansion include IL-12-producing monocytes and HLA-E (PMID 25384219).
- Adaptive-like features can be **engineered rather than waited for: Cichocki et al. (2021, PMID 34525347)** harnessed adaptive NK features (including FcεRIγ-low/CD16-driven signalling) to build iPSC-derived NK cells with enhanced antibody-dependent function.
- Practical caveat: adaptive expansions depend on donor CMV serostatus, making them a **donor-selection** lever for primary products and a **design target** for iPSC products.

Priming state changes what a subset can do

Wagner et al. (2017, PMID 28972539) showed CD56bright NK cells — usually written off as non-cytotoxic — mount potent antitumour responses after IL-15 priming. Combined with the observation that distinct developmental pathways generate functionally distinct NK populations (PMID 38872000), the operative variable is state at the time of infusion, which manufacturing controls directly.

Dysfunction, exhaustion and plasticity

The product's phenotype does not survive contact with the tumour untouched:

- **NK→ILC1 conversion.** Gao et al. (2017, PMID 28759001) showed TGF-β-dependent conversion of effector NK cells into intermediate ILC1-like cells as a tumour immunoevasion mechanism; Cxcr3-driven infiltration also produces ILC1-like plasticity in liver metastasis (PMID 40168086).
- **Hypoxia.** Ni et al. (2020, PMID 32445619) profiled tumour-infiltrating NK cells by scRNA-seq and showed that deleting Hif1a unleashes NK activity; hypoxia impairs cytotoxicity via SHP-1/STAT3/ERK (PMID 33123602) and via autophagic granzyme B degradation in the target cell (PMID 24101526).
- **Intratumoral phenotype collapse.** Platonova et al. (2011, PMID 21708957) documented coordinated alterations of intratumoral NK phenotype and function in NSCLC; more recently, NK cells become functionally impaired within hours of tumour entry (PMID 38267402).
- **Checkpoint and epigenetic dysfunction.** High NKG2A marks exhaustion and poor prognosis in liver cancer (PMID 28197391); a PD-1-high NK subset exists in patients (PMID 27372564); BATF drives epigenetic dysfunction of NK cells in AML (PMID 39259809); NKG2A overexpression drives exhaustion in relapsed AML (PMID 40320412).

- **Reversibility.** Retinoic-acid-receptor activation reprogrammes the senescence response and improves NK antitumour activity (PMID 38428412); fasting reshapes tissue niches to improve NK-mediated immunity (PMID 38878769) — evidence that the dysfunctional state is pharmacologically addressable.

Education, KIR and the donor-selection layer

Education/licensing still predicts clinical outcome: KIR B-haplotype donor selection improves survival after unrelated transplant for AML (PMID 20581313), the **HLA-B –21 M/T dimorphism** shapes NK education and immunotherapy outcome in AML (PMID 30647027), and DNAM-1/KIR receptor ratios have been proposed as predictive biomarkers in solid tumours (PMID 30242020). For allogeneic products this is a cheap efficacy lever available before any engineering.

Summary — the phenotype checklist for a modern product

Lever	Best-supported implementation
Differentiation state	CIML (IL-12/15/18, 12–16 h) or IL-15-primed
Adaptive features	CMV+ donor selection, or engineered FcεRIγ-low/adaptive-like iPSC-NK
Education	KIR B-haplotype / HLA-B –21 donor selection; KIR-ligand mismatch
Resistance to reprogramming	TGF-β/SMAD-resistant, HIF-1α-independent, metabolically armoured
Characterisation	scRNA-seq/CITE-seq or high-parameter flow of the final product, not the starting apheresis

Cell sources: blood, cord blood, iPSC and cell lines

Source choice fixes the ceiling on expansion, editability, homogeneity and scale, and it interacts with every downstream manufacturing decision.

Peripheral blood (PB-NK)

- Haploidentical PB-NK infusions established the modern field: **Miller et al. (2005, PMID 15632206)** demonstrated safe adoptive transfer with in vivo expansion after lymphodepletion, following on **Ruggeri et al. (2002, PMID 11896281)**, which showed donor-versus-recipient NK alloreactivity eliminates AML relapse and graft rejection without GvHD.
- Strengths: mature, educated, fully cytotoxic cells; KIR/HLA donor selection possible.
- Limits: low starting frequency (~10% of lymphocytes), donor-to-donor variability, one-donor-one-patient economics unless expansion is very high, and T-cell depletion required for allogeneic use.
- Patient-derived (autologous) NK cells are frequently dysfunctional — reduced CD16+ cytotoxic NK cells in myeloma marrow predict worse daratumumab responses (PMID 40019438), and NK cells from cancer patients show diminished cytotoxicity versus healthy donors (PMID 20849361), which is the core argument for allogeneic products.

Cord blood (CB-NK)

- The dominant substrate for engineered clinical products: cord blood supports lentiviral transduction plus IL-15 armouring and log-scale expansion, and units are banked, HLA-typed and immediately available (Liu 2018, PMID 28725044 → the CD19 CAR-NK trials, PMID 32023374 and PMID 38238616).
- **Shah et al. (2013, PMID 24204673)** achieved ~1,800–2,400-fold expansion from fresh or cryopreserved CB using aAPC feeders in gas-permeable vessels — demonstrating that a single unit can yield multiple doses.
- CB-NK are more immature (higher CD56bright fraction, lower KIR), which is partly compensated by cytokine preactivation and armouring, and CB-derived TRACK NK cells (PD-L1+, sIL-15-secreting) have reached the clinic in NSCLC (PMID 38572955; PMID 39903538).
- Selective-thaw devices allow a small aliquot of a banked unit to be used for NK expansion while preserving the unit for transplant (PMID 26432560).

iPSC-derived NK cells (iNK)

The route to a genuinely homogeneous, multi-edited, off-the-shelf product:

- **Knorr et al. (2013, PMID 23515118)** established clinical-scale derivation of NK cells from human pluripotent stem cells; **Zhu & Kaufman (2019, PMID 31396935)** improved the process to clinical scale without stromal feeders.

- **Li et al. (2018, PMID 30082067)** engineered iPSC-NK cells with an NK-optimised CAR (NKG2D TM + 2B4 + CD3 ζ) that outperformed a CAR-T-derived architecture in NK cells.
- Multiplex editing is where iNK cells differentiate themselves: CISH knockout improves persistence and IL-15-independent function (Zhu 2020, PMID 32531207); TGF- β pathway disruption is required for effective killing of hepatocellular carcinoma (PMID 38986609); high-affinity non-cleavable CD16a improves ADCC (PMID 31856277); adaptive-NK features can be built in (PMID 34525347); synNotch programming converts TIGIT and CD73 into activating inputs for glioblastoma (PMID 38429294).
- Differentiation protocol matters: two iPSC-NK differentiation strategies produce distinct signatures and cytotoxicity (PMID 39445004) — a manufacturing choice with a phenotypic readout.
- Clinical proof of principle: **FT596** (CD19 CAR + hnCD16 + IL-15/IL-15R α fusion) in B-cell lymphoma, phase 1 (PMID 39798981).

NK-92 and other cell lines

- NK-92 (Gong 1994, PMID 8152260) offers unlimited, consistent supply; infusions are safe in phase I (Arai 2008, PMID 18836917; Tonn 2013, PMID 24094496) but the cells are irradiated before infusion, so they cannot persist — potency must come from dose and frequency.
- Engineered NK-92 remains clinically relevant: ErbB2/HER2-CAR NK-92 (NK-92/5.28.z) went from GMP-compliant clonal line (PMID 25373520) to intracranial administration in recurrent glioblastoma (PMID 37148198).
- Line-specific improvements: CRISPR RNP editing enhances NK-92 cytotoxicity (PMID 32528479); expansion and cryopreservation workflows for clinical NK-92 manufacture are now published (PMID 38394177).
- Trade-off summary: unlimited scale and cheap consistency, at the cost of persistence, an aneuploid EBV+ lymphoma-derived genome and mandatory irradiation.

Making allogeneic sources truly universal

- **HLA class I knockout** converts allogeneic primary NK cells into off-the-shelf effectors by removing host-T-cell recognition (PMID 33584651); the mirror-image problem — host NK rejection of HLA-null cells — is why HLA-E/single-chain-HLA retention strategies appear in iPSC products.
- Placental/decidual NK cells (PMID 23641243) and CD34+ progenitor-derived NK cells widen the donor pool; microtransplantation delivers G-CSF-mobilised donor mononuclear cells containing large NK numbers (PMID 42035877).

Choosing a source

	PB-NK	CB-NK	iPSC-NK	NK-92
Doses per donor/lot	low-medium	medium-high	effectively unlimited	unlimited
Editability	moderate	moderate-good	best (clonal, multiplex)	good
			weeks when armoured	none (irradiated)

	PB-NK	CB-NK	iPSC-NK	NK-92
Persistence in vivo	days-weeks (with IL-15/lymphodepletion)	weeks when IL-15-armoured		
Maturity/education	mature, educated	immature, needs priming	protocol-dependent	line-fixed
Main risk	variability, T-cell contamination	immaturity	differentiation drift, tumourigenicity testing	no persistence, genomic background

CAR-NK and receptor engineering

NK cells need NK signalling modules

The single most transferable design lesson is that CAR-T architectures do not port cleanly:

- **Li et al. (2018, PMID 30082067)** screened CAR constructs in iPSC-NK cells and found a CAR combining the **NKG2D transmembrane domain, 2B4 costimulatory domain and CD3ζ** gave the strongest antigen-specific NK signalling and outperformed the standard CD28-CD3ζ CAR-T design in NK cells.
- DAP10/DAP12-based two-chain receptors exploit native NK adaptors; a KIRS2/DAP12 CAR improved safety and efficacy in r/r B-ALL (PMID 34703879), showing the concept is clinically viable.
- Counterintuitively, **CD28 — absent from mature NK cells — is a potent NK costimulatory module**: a CD70-CAR with a CD28 domain augmented signalling through an LCK/CD3ζ/ZAP70 axis (PMID 38900051). The lesson is that costimulatory domains must be screened empirically in NK cells, not inherited from T-cell dogma.
- Integration site is a design variable: knocking ζ-deficient CARs into the endogenous CD3ζ locus conveys potent cytotoxicity in both T and NK cells (PMID 38493479).

Reviews that map the design space usefully: Gong et al. (PMID 33933160) on CAR-NK design and engineering, Xie et al. (PMID 32853984) on the CAR-NK rationale, and the receptor-engineering review at PMID 38443448.

Targets and disease context

- Haematological: CD19 (PMID 32023374, PMID 38238616, PMID 40251398), CD33 (PMID 39349459), CD38 (PMID 33375774, PMID 35135865), CS1/SLAMF7 in myeloma (PMID 24067492), CD123, CD70.
- Solid tumours: HER2/ErbB2 (PMID 25373520 → glioblastoma trial PMID 37148198), EGFR and EGFRvIII in GBM (PMID 26155832), mesothelin (PMID 40025022; CAR-CIML NK in ovarian cancer, PMID 38996027), NKG2D-ligand targeting (PMID 30396908 in T cells, informing NK use).
- Solid-tumour CAR-NK reviews: PMID 38245520, PMID 39134804 — both converge on trafficking, persistence and TME suppression rather than antigen choice as the limiting factors.

Failure modes discovered by engineering

- **Trogocytosis-driven fratricide and antigen loss.** CAR activation transfers cognate antigen from tumour to NK cell, lowering tumour antigen density and causing CAR-NK cells to kill each other; a **KIR-based inhibitory CAR** rescues both problems (Li et al. 2022, PMID 36175679). This is arguably the most important NK-specific CAR failure mode described to date.
- **Self-antigen fratricide.** CD38-CAR NK cells kill each other because NK cells upregulate CD38 during expansion; CD38 knockout plus an affinity-optimised CAR

resolves it (PMID 33375774), and CD38 deletion also eliminates daratumumab-induced fratricide (PMID 32603414).

- **Loss of metabolic fitness, not antigen escape, drives relapse.** Single-cell profiling of CAR-NK cells after adoptive transfer showed relapse coincides with metabolic collapse, reversible by cytokine engineering (PMID 37494448).

Gene delivery into NK cells

NK cells are refractory to conventional transduction; delivery method is a manufacturing-critical choice.

Method	Evidence	Notes
Baboon envelope pseudotyped lentivirus (BaEV)	CD19-BBz CAR-NK phase 1 (PMID 40251398)	High efficiency in primary/ CB NK without feeders
Alpharetroviral / lentiviral	PMID 32117200	Direct comparison for CD19 CAR-NK in ALL
Retroviral	TRACK NK sIL-15 (PMID 38572955)	Used in clinical CB-NK products
Electroporation of plasmid/ mRNA	PMID 31114587	Non-viral, transient; CAR + CCR7 co-delivery
CRISPR RNP (Cas9 protein)	PMID 32528479, PMID 31704085	Avoids DNA toxicity; up to ~90% editing of primary NK
AAV-SB (Sleeping Beauty)	in vivo CRISPR screens in primary NK (PMID 38918616)	Screening platform, also a delivery template
Site-specific knock-in	CD38 locus knock-in of high-affinity CD16 (PMID 35135865); CD3ζ locus (PMID 38493479)	Combines knockout and knock-in in one edit

Feeders interact with transduction efficiency: feeder-based activation raises both proliferation and transduction (PMID 35222382, PMID 28810809, PMID 40325497), so delivery method and expansion platform must be co-optimised rather than chosen independently.

Synthetic-biology layer (mostly preclinical)

- **Inhibitory/logic-gated CARs:** KIR-based iCAR against trogocytosis (PMID 36175679); synNotch-programmed iPSC-NK cells that convert TIGIT/CD73 engagement into activation in glioblastoma (PMID 38429294).
- **Inducible switches:** rimiducid-controlled MyD88/CD40 (iMC) plus IL-15 enhances CAR-NK growth and cytotoxicity (PMID 32384544).
- **Cytokine-secreting "armoured" designs:** IL-15 (PMID 28725044), neoleukin-2/15 mutein driving c-Myc/NRF1 (PMID 40025022), decoy-resistant IL-18 delivered in trans by engineered bacteria (PMID 39367093), oncolytic-virus-delivered IL-15/IL-15Rα combined with EGFR-CAR NK cells.

- **Engager-secreting effectors:** CAR-T cells that secrete BiKEs to recruit endogenous NK cells (PMID 40659834) — an explicit bet that combining effector classes beats optimising one.

Design checklist

1. Use NK-native signalling modules (NKG2D-TM/2B4/DAP10/DAP12) **and** empirically test non-native costimulation (CD28, 4-1BB).
2. Anticipate trogocytosis and self-antigen fratricide; plan an inhibitory CAR or a knockout of the target on the effector.
3. Co-express or supply IL-15 — an unarmoured CAR-NK product will not persist.
4. Choose the delivery method with the expansion platform, not after it.
5. Assume the TME will suppress the product: add TGF- β resistance, checkpoint deletion or metabolic armouring in the same construct generation.

NK-cell engagers, CD16 biology and ADCC combinations

Engagers supply tumour specificity without gene transfer, which makes them the cheapest way to redirect an otherwise unmodified (and therefore easily manufactured) NK product.

BiKEs → TriKEs → multispecific NKCEs

- **BiKE (CD16 × CD33)** activated NK cells against primary MDS blasts and MDSC targets (Gleason 2014, PMID 24652987) — proof that crosslinking CD16 to a tumour antigen is sufficient for NK redirection.
- **TriKE** adds an IL-15 moiety to the same scaffold: the 161533 (CD16 × IL-15 × CD33) TriKE conferred specificity **plus** persistence, in vivo expansion and enhanced function (Vallera 2016, PMID 26847056). This is the key conceptual step — the engager also becomes the cytokine support.
- Target expansion: CD19-TriKE restores NK function in CLL (PMID 30890546), CLEC12A-TriKE targets AML blasts and leukaemic stem cells (PMID 33097838), and an IL-15-modified **NKp30 × HER2** TriKE avoids reliance on CD16 altogether (PMID 40694640).
- **Beyond CD16**: a trifunctional NKp46 × CD16a × CD123 NKCE controlled AML in vivo and overcame CD64-mediated resistance to anti-CD123 ADCC (Gauthier 2023, PMID 36635380). Engaging NKp46 sidesteps CD16 polymorphism and shedding.
- Construction methods are published as protocols (PMID 27177679), and reviews map the class (PMID 37638058).

AFM13: the engager that carries a cell product

AFM13 (CD30 × CD16A) is the clearest example of engager-product co-formulation:

- Pre-complexing AFM13 with cytokine-activated blood- or cord-blood-derived NK cells produced "CAR-like" responses against CD30+ malignancies without gene transfer (Kerbaui 2021, PMID 33986022).
- That combination reached the clinic: **cord-blood-derived, cytokine-preactivated, expanded NK cells pre-complexed with AFM13** in CD30+ lymphoma refractory to brentuximab vedotin and checkpoint inhibitors (phase 1, PMID 40186077) — one of the strongest recent efficacy signals for a non-CAR NK product.
- Manufacturing consequence: engager-loaded NK cells must survive cryopreservation as a complex, and this has been shown to be feasible with retained activity and specificity (PMID 35225870).

CD16 (FcγRIIIa) is the limiting component of ADCC

- **Shedding.** Activation induces ADAM17-mediated CD16A cleavage as negative feedback. Blocking ADAM17 enhances IL-15-mediated NK activation and proliferation (PMID 39053944), CD16A shedding limits engager-induced **serial killing** (PMID 41116262), and ADAM17 knockout is an established edit (PMID 31704085).

- **Non-cleavable, high-affinity CD16.** iPSC-NK cells expressing high-affinity non-cleavable CD16a (hnCD16) mediate improved ADCC (PMID 31856277) and this module is part of the clinically tested FT596 product (PMID 39798981). Alternatively, high-affinity CD16-158V can be knocked into the CD38 locus while simultaneously deleting CD38 (PMID 35135865).
- **Polymorphism.** The FcγRIIIA L48-H/R polymorphism enhances ADCC by promoting serial killing (PMID 39666369) — relevant both to donor selection and to which CD16 variant an engineered product should carry.

Antibody and drug combinations

- **Anti-CD38 (daratumumab) + expanded NK cells** in myeloma: effective ex vivo against patient cells (PMID 33457074), but daratumumab depletes CD38-high NK cells — hence CD38 knockout products (PMID 32603414, PMID 33375774). Epigenetic control of CD38/CD48 by KDM6A modulates the NK response (PMID 38355622), and the marrow NK profile predicts MRD negativity on daratumumab-based therapy (PMID 40019438).
- **Rituximab:** T-cell help in the TME enhances rituximab-mediated NK ADCC (PMID 38457360); the SUMOylation inhibitor subasumstat potentiates rituximab activity via IFN-I-dependent macrophage and NK stimulation (PMID 35226739).
- **Cetuximab:** CTLA-4+ Tregs expand in cetuximab-treated head-and-neck cancer and suppress NK cytotoxicity, correlating with poor prognosis (PMID 25832655) — a rationale for the CTLA-4 blockade arm in the CIML NK head-and-neck trial (PMID 39948608). Autologous NK cells (SNK01) plus chemotherapy and/or cetuximab were tested in NSCLC after TKI failure (PMID 38538093).
- **Radiotherapy** orchestrates NK-dependent responses through CXCL8 (PMID 35319989) and enhances NK cytotoxicity and tumour localisation, including a first-in-dog clinical trial (PMID 29254507).
- **Proteasome/epigenetic sensitisation of the target:** bortezomib suppresses HLA-E and upregulates DR5, sensitising myeloma to NK cells (PMID 30713790); BET inhibition activates NK cells via BRD4/SMAD3 in NSCLC (PMID 38519469); STING agonists enhance CAR-NK activity in pancreatic cancer (PMID 35371622).

Practical implications

1. If the product is unmodified NK cells, an IL-15-containing TriKE gives specificity **and** persistence in one reagent — an attractive alternative to CAR engineering.
2. Protect CD16: ADAM17 blockade or knockout, or express hnCD16.
3. For antibody combinations, check whether the antibody target is also expressed on NK cells (CD38 is the cautionary example) and delete it if so.
4. Engager pre-complexing is compatible with cryopreserved, off-the-shelf logistics.

Gene editing: removing the brakes

Adding a receptor gives specificity; deleting a brake gives fitness. The 2020–2026 literature has converged on a fairly short, well-replicated edit list, and unbiased screens are now expanding it.

The cytokine-signalling brake: CISH

- **Zhu et al. (2020, PMID 32531207)** deleted CISH in iPSC-NK cells: increased IL-15-driven JAK/STAT signalling, better expansion, greater cytotoxicity at low cytokine concentrations and improved in vivo persistence — i.e. the edit reduces the product's dependence on exogenous IL-15.
- **Daher et al. (2021, PMID 32902645)** combined CISH knockout with IL-15-armoured ("fourth-generation") CAR engineering of cord-blood NK cells, boosting effector function through the Akt/mTORC1 axis.
- A conditional Cish-deficient mouse model confirmed enhanced natural-cytotoxicity-receptor signalling and reduced NK exhaustion in solid tumours (PMID 35589278).

The HLA-E brake: NKG2A / KLRC1

- Anti-NKG2A antibody (monalizumab) unleashes both NK and CD8 T cells (André 2018, PMID 30503213), and NKG2A is a therapeutic vulnerability in MHC-I-heterogeneous triple-negative breast cancer (PMID 37791898).
- Genetic deletion is more attractive for a cell product: KLRC1 knockout improves NK activity against solid tumours (PMID 37675109); CRISPR editing of NKG2A improves primary **CD33-CAR NK** efficacy against AML (PMID 39349459); NKG2A knockout enhances cytotoxicity against myeloma (PMID 35694192).
- Tumours actively exploit this axis — HLA-E:NKG2A mediates escape of circulating tumour cells from NK surveillance (PMID 36706761).

The TGF- β / activin brake

- TGF- β represses mTOR in NK cells (PMID 26884601), drives metabolic dysfunction in human metastatic breast cancer (PMID 33568351), drives NK $\rightarrow$ ILC1 conversion (PMID 28759001) and acts through SMAD3 to suppress NK development via E4BP4/NFIL3 (PMID 28262747).
- Edits: **SMAD4 knockout** makes human NK cells resistant to both TGF- β and activin A while retaining cytotoxicity and IL-2/IL-15-driven proliferation, with better tumour penetration (PMID 40119192); TGF- β -pathway disruption is required for iPSC-NK efficacy against hepatocellular carcinoma (PMID 38986609); dominant-negative TGFBR2 and α v-integrin/TGF- β axis targeting improve activity against glioblastoma stem cells (PMID 34138753).

Fratricide and antibody-combination edits

- **CD38 knockout** removes daratumumab-induced fratricide and boosts effector activity (PMID 32603414), enables CD38-CAR NK cells (PMID 33375774), and its locus can host a high-affinity CD16 knock-in (PMID 35135865).
- **ADAM17 knockout** prevents CD16 shedding (PMID 31704085).
- **HLA class I knockout** makes allogeneic primary NK cells universal donors (PMID 33584651).

Newer intracellular targets

- **Regnase-1 / ZC3H12A** deletion increases IFN- γ via OCT2-dependent transcription and intratumoral accumulation (PMID 38821052).
- **Cbl-b** inactivation licenses NK cells to reject metastases, with TAM receptors as the upstream axis (Paolino 2014, PMID 24553136).
- **TIPE2** deletion improves adoptively transferred NK cells against solid tumours (PMID 36725083).
- **TIM-3** knockout enhances activity against glioma (PMID 33800561); TIM-3 ligand context determines whether this helps (PMID 39773563).
- **PD-1 (PDCD1)** knockout is feasible in primary NK cells (PMID 31704085); TIGIT blockade prevents NK exhaustion (Zhang 2018, PMID 29915296).

Unbiased screens are now the source of new targets

- **Genome-wide CRISPR screens in primary human NK cells** identified MED12, ARIH2 and other checkpoints whose ablation increases CAR-NK potency under immunosuppressive pressure (PMID 40845844).
- **In vivo AAV-SB-CRISPR screens of tumour-infiltrating NK cells** across four solid-tumour models identified genetic checkpoints of CAR-NK therapy alongside single-cell characterisation of TINK subpopulations (PMID 38918616).
- **Two-cell-type screens** shift the question to the target cell: CHMP2A loss in glioblastoma stem cells and HNSCC increases sensitivity to NK killing (PMID 35393416) — a druggable-sensitiser axis rather than an effector edit.
- Epigenetic regulators recur: EZH2 in NK differentiation (PMID 26668377), BATF in AML-driven NK dysfunction (PMID 39259809), DNMT1 inhibition improving CIML NK activity via autophagy (PMID 39704855), m6A methylation downstream of mTOR (PMID 38640466).

Editing platform and safety

- Cas9 RNP electroporation is the workhorse: up to ~90% editing of primary peripheral-blood NK cells (PMID 31704085), integrated with feeder-free expansion of cryopreserved NK cells (PMID 33433623), and applied to NK-92 (PMID 32528479).
- Feeder-based activation improves both expansion and editability (PMID 35222382).
- Multiplex editing raises translocation, off-target and clonality questions; **safety-assessment platforms for CRISPR-modified NK cells** are being formalised (PMID

41425600), and this — not editing efficiency — is now the rate-limiting step for products carrying 3+ edits.

Consensus edit stack for a solid-tumour product

1. CISH (cytokine-signalling brake) — persistence and IL-15 independence.
2. KLRC1/NKG2A — resist HLA-E-mediated inhibition.
3. TGF- β resistance (SMAD4 knockout or dnTGFBR2) — resist TME reprogramming.
4. ADAM17 knockout or hnCD16 knock-in — preserve ADCC.
5. Target-antigen knockout when the antigen is shared with NK cells (e.g. CD38).
6. Optional/emerging: Regnase-1, TIPE2, Cbl-b, MED12/ARIH2 from screens.

Cytokine support and armouring

Cytokines act at three separate points — ex vivo expansion, ex vivo differentiation (state programming), and in vivo support after infusion — and the literature is clearest when those are kept apart.

IL-2 versus IL-15

- IL-15 activates human NK cells through IL-2R components (Carson 1994, PMID 7523571) and IL-15 trans-presentation drives NK development and differentiation in vivo (PMID 19103877).
- Clinically, systemic IL-2 supports infused NK expansion (Miller 2005, PMID 15632206) but also expands regulatory T cells; depleting Tregs with an IL-2-diphtheria-toxin fusion raised the in vivo NK expansion rate from 10% to 27% of patients and improved AML clearance (Bachanova 2014, PMID 24719405).
- The field has therefore moved to **IL-15-based support** (IL-15 super-agonist N-803, IL-15/IL-15R α -Fc) or to **cell-intrinsic IL-15** so no systemic cytokine is needed.

Preactivation: IL-12 + IL-15 + IL-18 (CIML)

Covered in detail in `01_phenotypes_and_cell_states.md`. Manufacturing-relevant points:

- The stimulus is **brief** (12–16 h) and therefore easy to insert at the end of an existing process (Romee 2012, PMID 22983442; 2016, PMID 27655849).
- It is compatible with feeder expansion (PM21 particles + cytokines, PMID 38711506), with engager pre-complexing (PMID 33986022) and with CAR transduction (PMID 38996027).
- IL-27 augments IL-15/IL-18-mediated activation (PMID 31277710), and DNMT1 inhibition further improves CIML activity (PMID 39704855) — both examples of tuning the priming cocktail rather than replacing it.

IL-21: the expansion cytokine

IL-21 is used less for effector programming than for **proliferative capacity**, almost always as a membrane-bound feeder ligand (see `07_expansion_and_gmp_manufacturing.md`):

- Membrane-bound IL-21 aAPCs support >10,000-fold expansion over 5 weeks with retained cytotoxicity and increased metabolic activation (PMID 31624330; PMID 21339714).
- Short soluble IL-21 exposure combined with K562-OX40L feeders is an alternative (PMID 31105701).
- IL-21 with autologous feeder cells in a CliniMACS Prodigy workflow supports clinical-grade CAR-NK generation (PMID 28810809).

Cytokine armouring of the product

The dominant strategy for persistence without systemic cytokine toxicity:

Design	Evidence
Membrane-bound / intracellular IL-15 in NK-92	PMID 22310931 — IL-15 transduction enriches gene-modified effectors and removes IL-2 dependence
Secreted IL-15 + CD19 CAR in cord-blood NK	Liu 2018, PMID 28725044 → clinical (PMID 32023374, PMID 38238616)
IL-15/IL-15Rα fusion in iPSC-NK (FT596)	PMID 39798981
Soluble IL-15 in cord-blood TRACK NK	PMID 38572955; interim clinical report PMID 39903538
IL-15 + CISH knockout	PMID 32902645, PMID 32531207
Neoleukin-2/15 mutein secretion (c-Myc/NRF1)	PMID 40025022
Rimiducid-inducible MyD88/CD40 + IL-15	PMID 32384544
IL-15 in the engager rather than the cell (TriKE)	PMID 26847056, PMID 40694640
Decoy-resistant IL-18 delivered in trans	PMID 39367093
IL-15/IL-15Rα from an oncolytic virus + CAR-NK	OV-IL15C + EGFR-CAR NK
IL-15/4-1BBL ex vivo activation before infusion	PMID 25452614 — note: associated with acute GvHD after T-cell-depleted HSCT

Two cautions run through this table:

1. **IL-15 armouring is not free.** The IL-15/4-1BBL-activated NK trial reported acute GvHD after T-cell depleted transplant (PMID 25452614), and unregulated IL-15 raises theoretical risks of NK-cell autonomy; inducible or membrane-restricted designs exist for that reason.
2. **Armouring compensates for, but does not replace, metabolic fitness.** CAR-NK relapse was driven by metabolic collapse and was rescued by cytokine engineering (PMID 37494448) — the two axes are coupled (see `08_metabolism_persistence_trafficking.md`).

Practical schedule for a modern process

1. **Expansion phase:** IL-2 or IL-15 with membrane-bound IL-21/4-1BBL feeder or particle stimulation.
2. **Programming phase (final 12-16 h):** IL-12 + IL-15 + IL-18 to install the memory-like state.
3. **In vivo support:** cell-intrinsic IL-15 (secreted, membrane-bound or fusion) and/or an IL-15-containing engager; if systemic cytokine is used, prefer IL-15 super-agonist over IL-2 and consider Treg depletion.

Ex vivo expansion, feeders, media and GMP manufacturing

Dose is the binding constraint on NK therapy: NK cells are ~10% of blood lymphocytes, and clinical activity has repeatedly tracked with cell dose and in vivo expansion. This axis is where the process-development literature is densest (30 papers, 12 of them explicit process/method studies, plus the cryopreservation set in 09_cryopreservation_potency_and_release.md).

Feeder-based expansion: the K562 engineering lineage

Feeder	Ligands	Reported expansion	Source
K562-mb15-41BBL	membrane IL-15 + 4-1BBL	median 21.6-fold NK in 7 days (range 5.1–86.6, n=50)	Imai/Fujisaki 2009, PMID 19383914
K562-mb15-41BBL in G-Rex	as above, gas-permeable static culture	up to 19×10^9 NK cells in 8–10 days from unseparated apheresis	Lapteva 2012, PMID 22900959
K562-mbIL21-41BBL (aAPC)	membrane IL-21 + 4-1BBL	~1,848-fold (fresh CB) / ~2,389-fold (cryopreserved CB) in 14 days, >95% purity	Shah 2013, PMID 24204673
"NKF" mbIL-21 feeder line	membrane IL-21	>10,000-fold at 5 weeks, increased metabolic activation	PMID 31624330
K562-OX40L + short soluble IL-21	OX40L	robust expansion via OX40–OX40L axis	PMID 31105701
Autologous feeder cells + IL-21 (Prodigy)	—	clinical-grade NK and anti-CD123 CAR-NK	PMID 28810809
Engineered feeders displaying costimulatory molecules + IL-21	screened combinations	improved expansion and transduction	PMID 40325497
EBV-LCL feeders	—	clinical-grade cytotoxic NK with upregulated activating receptors and death-receptor ligands	PMID 19308771

Why membrane-bound presentation works: it delivers sustained, cell-contact-dependent signalling that soluble cytokines cannot reproduce, which is why the 4-1BBL/IL-21 combination has survived 15 years of process iteration. Feeders also **prime the cells for gene modification** — feeder-activated NK cells transduce and edit far better than resting cells (PMID 35222382).

Feeder liabilities: an irradiated tumour-derived cell line in the process requires residual-cell testing, irradiation validation and regulatory justification, plus a GMP feeder bank.

Feeder-free and cell-free routes

- **PM21 particles** — plasma-membrane particles derived from K562-mb21-41BBL — give feeder-like expansion (mean ~825-fold ex vivo) with in vivo biodistribution data (PMID

27059202), and with added cytokines produce memory-like characteristics and better survival (PMID 38711506). This is the most mature "feeder benefits without feeder cells" platform.

- **Automated feeder-free bioreactor expansion** of clinical-grade NK cells in a closed system was demonstrated early (PMID 20795758) and remains the reference for feeder-free GMP.
- **Feeder-free expansion integrated with non-viral genome editing** of cryopreserved primary NK cells (PMID 33433623) shows the two constraints can be met simultaneously.
- A 2025 review categorises feeder-free strategies (cytokine combinations, immobilised ligands, nanoparticle and bead-based aAPC mimics) and their trade-offs against feeder-based yield (PMID 41089682).

Media, serum and reagents

- **Direct comparison matters.** A 2025 Cytotherapy study screened RPMI 1640, KBM581, SCGM, NK MACS, X-VIVO 15 and AIM-V, each with FBS, human AB serum, human platelet lysate or Immune Cell Serum Replacement, plus feeders, and read out viability, fold expansion, cytotoxicity, immunophenotype and transcriptome (PMID 39570247). Medium/supplement choice changed both yield and phenotype — it is not a neutral variable.
- Serum-free/xeno-free operation is required for GMP; platelet lysate and defined serum replacements are the practical substitutes for AB serum.
- GMP-compliant large-scale protocols for alloreactive NK products in AML are published with full release data, giving concrete comparators for yield and purity.

Vessels, automation and scale

- **Gas-permeable static culture (G-Rex)** transformed achievable yields — 19×10^9 cells from one apheresis product in 8–10 days (PMID 22900959) — and is used for cord-blood aAPC expansion too (PMID 24204673).
- **CliniMACS Prodigy** provides fully automated selection→activation→expansion→transduction in one closed device (PMID 28810809); automated primary CAR-NK manufacture for AML has been demonstrated end-to-end (PMID 38253870).
- **Stirred/aerated bioreactors** are the current frontier: process development for scalable NK expansion in aerated stirred bioreactors moves the field from scale-out (many flasks) to scale-up (one vessel) (PMID 40808775), and scalable NK/CAR-NK expansion process development in bioreactor systems is now being reported (2024).
- **Clinical-scale iPSC-NK differentiation** is its own manufacturing problem — spin-EB/feeder-free differentiation at clinical scale (PMID 23515118, PMID 31396935), with differentiation strategy affecting the resulting phenotype and cytotoxicity (PMID 39445004).

What a modern NK manufacturing process looks like

1. **Starting material:** apheresis (CD3-depleted / CD56-selected), cord-blood unit, or iPSC master cell bank.

2. **Activation/expansion:** membrane IL-21/4-1BBL feeder cells or PM21 particles + IL-2/IL-15, in gas-permeable vessels or a closed automated system; 10–14 days typical, 100–2,000-fold expansion.
3. **Genetic modification (if any):** BaEV or lentiviral transduction, or Cas9 RNP electroporation, timed to the feeder-activation window.
4. **Programming:** terminal IL-12/15/18 preactivation for memory-like state.
5. **Harvest, formulate, cryopreserve:** DMSO-based or DMSO-free cryoprotectant, controlled-rate freeze.
6. **Release:** identity/purity (CD56+CD3–), viability, potency (cytotoxicity), residual feeder cells, vector copy number/editing efficiency, sterility, endotoxin, mycoplasma.

Open process questions

- No head-to-head, multi-site comparison of feeder versus PM21 versus feeder-free media platforms using the same donors and the same potency assay.
- Expansion fold and cytotoxicity are reported inconsistently (different E:T ratios, targets, assay formats), which makes cross-paper yield comparisons unreliable.
- Bioreactor NK expansion data remain thin relative to flask/bag data, particularly for edited or CAR-transduced cells.
- Cost of goods is almost never reported, despite being the practical argument for allogeneic and iPSC products.

Metabolic fitness, persistence, trafficking and the solid-tumour microenvironment

This is where NK therapy still fails, and where 2024–2026 work is concentrated.

Metabolism is a therapeutic axis, not a biomarker

- Cytokine-driven glycolysis and OXPHOS are required for NK effector function; amino-acid-dependent cMyc expression is the checkpoint (PMID 29904050).
- **Obesity** drives PPAR-dependent lipid accumulation that paralyzes NK metabolism and antitumour responses (Michelet 2018, PMID 30420624) — a patient-selection and pretreatment consideration.
- **TGF- β** represses mTOR (PMID 26884601) and drives NK metabolic dysfunction in human metastatic breast cancer (PMID 33568351).
- **Hypoxia** impairs cytotoxicity via SHP-1-mediated attenuation of STAT3/ERK (PMID 33123602) and via HIF-1 α -dependent transcriptional reprogramming, whose deletion unleashes tumour-infiltrating NK activity (PMID 32445619); IL-2 activation can partially override hypoxic inhibition in myeloma marrow (PMID 23724099).
- **Autophagy** is required for NK function — C/EBP β -dependent autophagy inhibition impairs NK cells in cancer (PMID 39609420) — while autophagy in the tumour cell degrades granzyme B and confers resistance (PMID 24101526).
- **Nutrient and metabolite competition:** vitamin B6 restriction by pancreatic cancer cells (PMID 37931287), glutamine deprivation via myCAF exosomal PWAR6 in colorectal liver metastasis (PMID 39696364), and lactate-driven lysine **lactylation** of NK proteins, which can be targeted to reinforce cytotoxicity (PMID 40494934).
- **Mitochondria:** the mitochondrial apoptosis pathway determines killing efficiency at physiological E:T ratios, and targeting it augments NK immunotherapy (Pan 2022, PMID 35447071); mitochondrial dynamics in the TME are reviewed at PMID 41075850.
- **Engineering response:** the clearest demonstration that this is actionable is CAR-NK relapse caused by loss of metabolic fitness and rescued by cytokine engineering (PMID 37494448); DHA supplementation (PMID 38526128) and fasting-induced niche remodelling (PMID 38878769) are non-genetic levers.

Manufacturing link: expansion conditions set the metabolic starting point. mbIL-21 feeder expansion increases metabolic activation (PMID 31624330), and cryopreservation depletes it (see 09_cryopreservation_potency_and_release.md), so "metabolic fitness" is partly a process attribute that could be released against.

Persistence in vivo

- Without support, infused NK cells persist days; with lymphodepletion plus cytokine support they expand measurably in a minority of patients (10% → 27% with Treg depletion, PMID 24719405).
- Cell-intrinsic IL-15 changes the picture: IL-15-armoured CD19 CAR-NK cells were detectable long-term in preclinical models (PMID 28725044) and CAR19/IL-15 NK persistence was a defined secondary endpoint in the phase 1/2 trial (PMID 38238616).

- CIML differentiation confers weeks-long enhanced responsiveness and measurable in vivo expansion in patients (PMID 35349491).
- Intrinsic survival programmes matter too: 2B4 signalling inhibits NK apoptosis via pERK/BCL-2 (PMID 37516949), and CISH/Regnase-1/TIPE2 edits all improve persistence or intratumoral accumulation.

Trafficking and infiltration

- **Chemokine receptor engineering** is the direct approach: CCR7 acquisition by trogocytosis redirects expanded NK cells to lymph nodes (PMID 22498742); CCR7 co-delivery with a CAR by electroporation (PMID 31114587); CXCR3 drives NK infiltration and plasticity in colorectal liver metastasis (PMID 40168086). Homing biology is reviewed comprehensively at PMID 35768424.
- **Physical and stromal barriers:** cancer-associated fibroblasts act as decoys that absorb NK cytotoxicity in breast cancer (PMID 40052789); melanoma-associated fibroblasts reshape NK phenotype (PMID 19934056); α v-integrin/TGF- β signalling limits activity against glioblastoma stem cells (PMID 34138753).
- **Suppressive cell networks:** Tregs inhibit NK cells in a TGF- β -dependent manner (PMID 16230475); MDSCs suppress NK cells via NKp30 in hepatocellular carcinoma (PMID 19551844); tumour-associated NK cells can themselves drive MDSC-mediated tolerance through IL-6/STAT3 (PMID 38748775) — a reminder that infused NK cells are not unconditionally beneficial.
- **Rapid functional loss on entry:** NK cells become impaired within hours of entering the tumour (PMID 38267402), which argues that trafficking without accompanying resistance engineering is insufficient.
- **The NK-dendritic-cell axis** is the main indirect mechanism of benefit: NK cells recruit cDC1 via chemokines (Böttcher 2018, PMID 29429633) and an NK-DC axis defines checkpoint-therapy-responsive tumours (Barry 2018, PMID 29942093). This supports combining NK products with T-cell-directed therapy rather than positioning them as alternatives.

Practical implications

1. Release-test metabolic/functional fitness (post-thaw 3D migration or serial-killing capacity), not just viability and single-round cytotoxicity.
2. Engineer for the microenvironment: TGF- β resistance, HIF-1 α /hypoxia tolerance, lactate/lactylation resistance, and where relevant a chemokine receptor matched to the target tissue.
3. For solid tumours, consider intratumoral/intracavitary administration — the HER2 CAR-NK glioblastoma trial injected cells into the resection cavity (PMID 37148198), bypassing trafficking entirely.
4. Expect and exploit the NK $\rightarrow$ DC $\rightarrow$ T-cell axis when designing combination regimens.

Cryopreservation, potency assays and product release

Off-the-shelf NK therapy requires cryopreserved product. The evidence says this is a **potency problem**, not a logistics problem — and that the standard release assays hide it.

Cryopreservation damages the functions that matter most

- **Mark et al. (2020, PMID 33067467)** is the pivotal result: standard degranulation and chromium-release assays confirmed that cryopreserved NK cells still kill, yet the same cells showed markedly impaired **3-D migration and cytotoxicity in tissue-like matrices** (earlier preprint of the same work at bioRxiv). A product that passes a 2-D potency assay can be functionally inert in tissue.
- Head-to-head comparison of fresh versus cryopreserved/thawed clinical-grade expanded NK cells confirms measurable functional loss (PMID 32536506).
- Mechanistic work in 2025–2026 localises the damage: lysosomal damage triggering programmed cell death, which induced stress granules can alleviate (PMID 42098076), and composition/temperature/granule-dependent post-thaw survival and function (PMID 42564981). A dedicated study of cryopreservation damage mechanisms in NK cells appeared in Cytotherapy (PMID 39918490).

Reducing or replacing DMSO

DMSO is both the standard cryoprotectant and a source of dose-dependent infusion toxicity.

- Systematic evaluation of DMSO/Me₂SO, proline, trehalose and dextran-40 combinations for NK cells (PMID 41208818).
- DMSO-free formulations benchmarked against CryoStor 10 and FBS+DMSO for recovery and function (PMID 42098076; PMID 42564981).
- **Intracellular nanoparticle-mediated protection** avoids cryoinjury while retaining antitumour function (PMID 32382476) — a route that changes the cell rather than the buffer.
- Post-thaw **rescue** rather than prevention: brief co-culture with effector T cells or synthetic cells restores motility and killing after thaw (PMID 40966444).

Cryopreservation of complex products

- Cord-blood-derived NK cells expanded from long-term cryopreserved units remain cytotoxic against primary breast cancer cells (PMID 29189387), and selective-thaw devices let a banked unit serve both transplant and NK manufacture (PMID 26432560).
- Cryopreserved NK cells retain activity against leukaemic targets, with IL-2 co-application affecting the result (PMID 27614454).
- **Engager-loaded cells can be frozen as a complex**: NK cells pre-complexed with innate cell engagers retain activity and specificity after cryopreservation (PMID 35225870) — the enabling result for off-the-shelf AFM13-NK logistics (PMID 40186077).

- Clinical NK-92 manufacturing has published expansion **and** cryopreservation conditions together (PMID 38394177), which is the right unit of process description.

Potency assays and release testing

- Conventional readouts — chromium/calcein release, CD107a degranulation, IFN- γ — are 2-D, short-duration and single-round, and demonstrably fail to detect cryopreservation-induced defects (PMID 33067467).
- Better-correlating alternatives appearing in the literature:
- **3-D spheroid microarrays** for high-throughput, high-content NK cytotoxicity (PMID 34290356), and 3-D colorectal spheroids where activated NK killing was independent of target PD-L1 (PMID 29632716).
- **Serial-killing** assays, which reveal CD16-shedding-dependent limits invisible to single-round assays (PMID 41116262, PMID 39666369).
- **Patient-derived organoid co-culture**, which shows hypoxia/TGF- β -driven, patient-specific diversification of NK activation programmes (PMID 41360426).
- Sensitive metabolic viability readouts (resazurin-based) validated for NK-derived products (PMID 39022723).
- High-dimensional phenotyping and transcriptional profiling of the final product are now reported alongside clinical outcomes (PMID 39948608, PMID 38238616), which points toward phenotype-based release criteria (memory-like signature, CD16 expression, exhaustion markers) rather than purity and viability alone.

Recommended release panel for a cryopreserved NK product

Attribute	Assay
Identity/purity	CD56+CD3– frequency; residual T cells; residual feeder cells
Viability	post-thaw viability and recovery (both, not just viability)
Potency (primary)	3-D or spheroid cytotoxicity post-thaw, at clinically plausible E:T
Potency (secondary)	serial-killing capacity; CD107a/IFN- γ after receptor restimulation
Fitness	post-thaw migration; metabolic readout
Phenotype	CD16 (and cleavage status), NKG2A/NKG2D/NCRs, memory-like and exhaustion markers
Genetic modification	vector copy number, transgene expression, editing efficiency, off-target/translocation assessment (PMID 41425600)
Safety	sterility, mycoplasma, endotoxin, residual feeder/irradiation controls

Open problems

- No consensus potency assay, which makes cross-product and cross-trial comparison of "NK activity" unreliable.
- Post-thaw functional loss is well documented but rarely reported in clinical papers, so its contribution to weak solid-tumour efficacy is unquantified.

- DMSO-free formulations are validated mostly for unmodified NK cells; edited and CAR-transduced products may behave differently.

Clinical translation and remaining gaps

What has been established in patients

Safety is the field's strongest claim. Across allogeneic, cord-blood, iPSC-derived, engineered and cell-line products, investigators consistently report no GvHD and little or no CRS/ICANS — including the CD19 CAR-NK studies (PMID 32023374; Marin 2024, PMID 38238616), iPSC-derived FT596 (PMID 39798981), 4-1BB-costimulated CD19 CAR-NK (PMID 40251398) and intracranial HER2 CAR-NK (PMID 37148198). This is the principal differentiator from CAR-T therapy and holds across 120+ CAR-NK trials catalogued in the 2025 clinical landscape review (PMID 40149002).

Efficacy is real where three conditions coincide — haematological target, armoured/preactivated product, adequate lymphodepletion:

Product	Setting	Key point
Haploidentical PB-NK + IL-2	AML, non-transplant	first demonstration of in vivo expansion and remissions (PMID 15632206); Treg depletion raised expansion 10%→27% and improved clearance (PMID 24719405)
KIR-ligand-mismatched NK	elderly high-risk AML	feasible, no GvHD, CR in active disease (PMID 21791425)
CB-derived CAR19/IL-15 NK	r/r CD19+ B-cell malignancies	11-patient phase 1/2 (PMID 32023374) then 37 patients with determinants of response (PMID 38238616)
CD19-BBz CAR-NK (BaEV-transduced CB)	r/r large B-cell lymphoma	repeat dosing feasible, phase 1 (PMID 40251398)
FT596 (iPSC; CD19 CAR + hnCD16 + IL-15/IL-15Rα)	B-cell lymphoma	first-in-human iPSC-derived CAR-NK (PMID 39798981)
Donor CIML NK	post-transplant relapse	expansion, persistence and efficacy characterised (PMID 35349491); original AML trial PMID 27655849
CIML NK + N-803 ± ipilimumab	r/r head and neck cancer	first-in-human in solid tumour after median 6 prior lines (PMID 39948608)
AFM13-precomplexed CB-NK	CD30+ lymphoma refractory to BV and checkpoint inhibitors	phase 1 with meaningful responses in a dismal-prognosis population (PMID 40186077)
NK-92 / CAR-NK-92	RCC, melanoma, advanced cancers, recurrent GBM	safe at high doses; irradiated so non-persistent (PMID 18836917, PMID 24094496, PMID 37148198)
TRACK NK (CB, sIL-15, PD-L1+)	NSCLC refractory to checkpoint inhibitors	interim engineered-NK trial report (PMID 39903538)
Autologous SNK01 + chemo/cetuximab	NSCLC post-TKI	phase I/IIa with preclinical companion (PMID 38538093)

Solid tumours remain largely unsolved. The exceptions are instructive: intracranial delivery bypasses trafficking (PMID 37148198), and the head-and-neck CIML trial combines state programming, cytokine support and checkpoint blockade (PMID 39948608). Both add something beyond "infuse more NK cells".

Contributions to efficacy, ranked by strength of clinical evidence

1. **IL-15 armouring or IL-15-containing engagers** — present in essentially every product with a clear response signal.
2. **Lymphodepletion plus host Treg control** — the oldest and best-quantified lever (PMID 24719405).
3. **Memory-like (IL-12/15/18) programming** — clinical expansion/persistence data in two settings.
4. **Donor/education selection (KIR B haplotype, HLA-B *21, KIR-ligand mismatch)** — outcome associations in transplant cohorts (PMID 20581313, PMID 30647027).
5. **Non-cleavable/high-affinity CD16 plus an antibody or engager** — mechanistically strong, now in clinical products (FT596, AFM13-NK).
6. **Brake knockouts (CISH, NKG2A, TGF- β pathway)** — compelling preclinical data, clinical readouts still immature.
7. **Route of administration for solid tumours** — one positive trial (intracavitary GBM).

Gaps this literature makes visible

- **No standard potency assay.** Cryopreservation-induced loss of 3-D function is invisible to conventional release assays (PMID 33067467), so "the product met specification" does not imply it was potent.
- **Persistence remains weeks at best.** Cell-intrinsic IL-15 helps; nothing yet produces the multi-month persistence CAR-T achieves, and repeat dosing (PMID 40251398) is the pragmatic answer.
- **Manufacturing platforms have never been compared head-to-head** on shared donors with a shared potency readout: feeder versus PM21 versus feeder-free; flask versus G-Rex versus bioreactor; DMSO versus DMSO-free.
- **Multi-edit products outpace their safety science.** Screens are generating edit lists faster than translocation/clonality assessment frameworks mature (PMID 41425600).
- **Solid-tumour trafficking and TME resistance are still the bottleneck** — hypoxia, TGF- β , lactylation, nutrient competition, CAF decoys and within-hours functional impairment on tumour entry (PMID 38267402).
- **Cost of goods and dose economics are essentially unreported**, despite being the main practical argument for allogeneic, iPSC and off-the-shelf approaches.
- **Reporting heterogeneity** — expansion fold, E:T ratios, targets and assay formats differ so widely that quantitative meta-analysis across the process literature is currently impossible.

What a maximally-supported product would look like today

Cord-blood- or iPSC-derived NK cells; expanded with membrane-bound IL-21/4-1BBL feeders or PM21 particles in a closed, gas-permeable or bioreactor system with serum-free medium; transduced (BaEV lentivirus) with an NK-optimised CAR plus IL-15; edited for CISH, KLRC1 and TGF- β resistance with hnCD16 retained; given a terminal IL-12/15/18 memory-like preactivation; cryopreserved in a DMSO-minimised formulation and released against a 3-D post-thaw potency assay; infused after lymphodepletion, with repeat dosing, an antibody or engager partner, and intratumoral delivery where anatomy permits.

Every element of that sentence is supported by at least one study in this collection; **no published product combines all of them**, which is a reasonable statement of where the field is.

NK cells for cancer — evidence listing

313 curated records, 1993–2026. 244 (77%) report expansion, culture or manufacturing content; 153 have open-access full text mirrored under `fulltext/` (marked FT).

Evidence mix — Clinical trials: 50, Human translational: 28, Process / manufacturing methods: 58, Preclinical (in vivo): 105, Mechanistic / in vitro: 33, Reviews: 39

Disease context — pan-cancer: 149, lymphoma_cll: 32, aml_mds: 32, gi: 22, hnscc_melanoma: 21, myeloma: 18, breast: 14, gbm_cns: 12, neuroblastoma_sarcoma: 11, ovarian: 10, gu: 9, all_peds_heme: 8, lung: 6

Cross-cutting NK-cell therapy landscape

22 papers · 1994–2024 · 17 with culture/manufacturing content

Clinical trials

- **2002** Uherek C et al. — Retargeting of natural killer-cell cytolytic activity to ErbB2-expressing cancer cells results in efficient and selective tumor cell destruction. *Blood*. [PMID 12149207](#)
- **2006** Roda JM et al. — Natural killer cells produce T cell-recruiting chemokines in response to antibody-coated tumor cells. *Cancer research*. [PMID 16397268](#)
- **2009** Mustjoki S et al. — Clonal expansion of T/NK-cells during tyrosine kinase inhibitor dasatinib therapy. *Leukemia*. [PMID 19295545](#)

Human translational

- **2021** Mathewson ND et al. — Inhibitory CD161 receptor identified in glioma-infiltrating T cells by single-cell analysis. *Cell*. [PMID 33592174](#)
- **2023** Tang F et al. — A pan-cancer single-cell panorama of human natural killer cells. *Cell*. [PMID 37607536](#)

Process / manufacturing methods

- **1998** Vitale M et al. — NKp44, a novel triggering surface molecule specifically expressed by activated natural killer cells, is involved in non-major histocompatibility complex-restricted tumor cell lysis. *The Journal of experimental medicine*. [PMID 9625766](#) FT
- **1999** Cantoni C et al. — NKp44, a triggering receptor involved in tumor cell lysis by activated human natural killer cells, is a novel member of the immunoglobulin superfamily. *The Journal of experimental medicine*. [PMID 10049942](#) FT
- **2012** Kimura H et al. — EBV-associated T/NK-cell lymphoproliferative diseases in nonimmunocompromised hosts: prospective analysis of 108 cases. *Blood*. [PMID 22096243](#)

Preclinical (in vivo)

- **1994** Kägi D et al. — Cytotoxicity mediated by T cells and natural killer cells is greatly impaired in perforin-deficient mice. *Nature*. [PMID 8164737](#)

- **2001** Diefenbach A et al. — Rae1 and H60 ligands of the NKG2D receptor stimulate tumour immunity. *Nature*. PMID 11557981
- **2005** Gasser S et al. — The DNA damage pathway regulates innate immune system ligands of the NKG2D receptor. *Nature*. PMID 15995699
- **2005** Palumbo JS et al. — Platelets and fibrin(ogen) increase metastatic potential by impeding natural killer cell-mediated elimination of tumor cells. *Blood*. PMID 15367435
- **2009** Cella M et al. — A human natural killer cell subset provides an innate source of IL-22 for mucosal immunity. *Nature*. PMID 18978771
- **2024** Colucci M et al. — Retinoic acid receptor activation reprograms senescence response and enhances anti-tumor activity of natural killer cells. *Cancer cell*. PMID 38428412 FT
- **2024** Kath J et al. — Integration of ζ -deficient CARs into the CD3 ζ gene conveys potent cytotoxicity in T and NK cells. *Blood*. PMID 38493479 FT
- **2024** Delconte RB et al. — Fasting reshapes tissue-specific niches to improve NK cell-mediated anti-tumor immunity. *Immunity*. PMID 38878769

Mechanistic / in vitro

- **2008** Caligiuri MA — Human natural killer cells. *Blood*. PMID 18650461
- **2024** Dean I et al. — Rapid functional impairment of natural killer cells following tumor entry limits anti-tumor immunity. *Nature communications*. PMID 38267402 FT

Reviews

- **2008** Vivier E et al. — Functions of natural killer cells. *Nature immunology*. PMID 18425107
- **2016** Guillerey C et al. — Targeting natural killer cells in cancer immunotherapy. *Nature immunology*. PMID 27540992
- **2024** Vivier E et al. — Natural killer cell therapies. *Nature*. PMID 38383621
- **2024** Page A et al. — Development of NK cell-based cancer immunotherapies through receptor engineering. *Cellular & molecular immunology*. PMID 38443448 FT

NK phenotypes & cell states (subsets, adaptive/memory-like, dysfunction)

26 papers · 2009–2024 · 15 with culture/manufacturing content

Clinical trials

- **2018** Guillaumon CF et al. — NK Cell Education in Tumor Immune Surveillance: DNAM-1/KIR Receptor Ratios as Predictive Biomarkers for Solid Tumor Outcome. *Cancer immunology research*. PMID 30242020
- **2019** Hallner A et al. — The HLA-B -21 dimorphism impacts on NK cell education and clinical outcome of immunotherapy in acute myeloid leukemia. *Blood*. PMID 30647027
- **2022** Shapiro RM et al. — Expansion, persistence, and efficacy of donor memory-like NK cells infused for posttransplant relapse. *The Journal of clinical investigation*. PMID 35349491 FT

Human translational

- **2017** Pesce S et al. — Identification of a subset of human natural killer cells expressing high levels of programmed death 1: A phenotypic and functional characterization. The Journal of allergy and clinical immunology. PMID 27372564
- **2021** Shaim H et al. — Targeting the α v integrin/TGF- β axis improves natural killer cell function against glioblastoma stem cells. The Journal of clinical investigation. PMID 34138753
- **2024** Netskar H et al. — Pan-cancer profiling of tumor-infiltrating natural killer cells through transcriptional reference mapping. Nature immunology. PMID 38956379 FT

Process / manufacturing methods

- **2009** Balsamo M et al. — Melanoma-associated fibroblasts modulate NK cell phenotype and antitumor cytotoxicity. Proceedings of the National Academy of Sciences of the United States of America. PMID 19934056
- **2014** Ali TH et al. — Enrichment of CD56(dim)KIR + CD57 + highly cytotoxic NK cells in tumour-infiltrated lymph nodes of melanoma patients. Nature communications. PMID 25472612 FT
- **2016** Cichocki F et al. — CD56dimCD57+NKG2C+ NK cell expansion is associated with reduced leukemia relapse after reduced intensity HCT. Leukemia. PMID 26416461

Preclinical (in vivo)

- **2011** Platonova S et al. — Profound coordinated alterations of intratumoral NK cell phenotype and function in lung carcinoma. Cancer research. PMID 21708957
- **2012** Romee R et al. — Cytokine activation induces human memory-like NK cells. Blood. PMID 22983442
- **2017** Gao Y et al. — Tumor immunoevasion by the conversion of effector NK cells into type 1 innate lymphoid cells. Nature immunology. PMID 28759001
- **2017** Wagner JA et al. — CD56bright NK cells exhibit potent antitumor responses following IL-15 priming. The Journal of clinical investigation. PMID 28972539
- **2017** Sun C et al. — High NKG2A expression contributes to NK cell exhaustion and predicts a poor prognosis of patients with liver cancer. Oncoimmunology. PMID 28197391
- **2020** Ni J et al. — Single-Cell RNA Sequencing of Tumor-Infiltrating NK Cells Reveals that Inhibition of Transcription Factor HIF-1 α Unleashes NK Cell Activity. Immunity. PMID 32445619
- **2021** Woan KV et al. — Harnessing features of adaptive NK cells to generate iPSC-derived NK cells for enhanced immunotherapy. Cell stem cell. PMID 34525347
- **2024** Tarannum M et al. — CAR memory-like NK cells targeting the membrane proximal domain of mesothelin demonstrate promising activity in ovarian cancer. Science advances. PMID 38996027 FT
- **2024** Ding Y et al. — Distinct developmental pathways generate functionally distinct populations of natural killer cells. Nature immunology. PMID 38872000

Mechanistic / in vitro

- **2013** Horowitz A et al. — Genetic and environmental determinants of human NK cell diversity revealed by mass cytometry. Science translational medicine. PMID 24154599
- **2017** Freud AG et al. — The Broad Spectrum of Human Natural Killer Cell Diversity. Immunity. PMID 29166586
- **2020** Dogra P et al. — Tissue Determinants of Human NK Cell Development, Function, and Residence. Cell. PMID 32059780
- **2024** Rebuffet L et al. — High-dimensional single-cell analysis of human natural killer cell heterogeneity. Nature immunology. PMID 38956378 FT
- **2024** Hofman T et al. — IFN γ mediates the resistance of tumor cells to distinct NK cell subsets. Journal for immunotherapy of cancer. PMID 38955423 FT
- **2024** Foltz JA et al. — Cytokines drive the formation of memory-like NK cell subsets via epigenetic rewiring and transcriptional regulation. Science immunology. PMID 38941480

Reviews

- **2017** Bi J et al. — NK Cell Exhaustion. Frontiers in immunology. PMID 28702032 FT
- **2018** Chiossone L et al. — Natural killer cells and other innate lymphoid cells in cancer. Nature reviews. Immunology. PMID 30209347

Cell sources: peripheral blood, cord blood, iPSC, cell lines

20 papers · 1994–2026 · 18 with culture/manufacturing content

Clinical trials

- **2013** Tonn T et al. — Treatment of patients with advanced cancer with the natural killer cell line NK-92. Cytotherapy. PMID 24094496
- **2016** Suck G et al. — NK-92: an 'off-the-shelf therapeutic' for adoptive natural killer cell-based cancer immunotherapy. Cancer immunology, immunotherapy : CII. PMID 26559813
- **2017** Zhang C et al. — Chimeric Antigen Receptor-Engineered NK-92 Cells: An Off-the-Shelf Cellular Therapeutic for Targeted Elimination of Cancer Cells and Induction of Protective Antitumor Immunity. Frontiers in immunology. PMID 28572802 FT
- **2024** Lu T et al. — Preclinical Evaluation of Off-The-Shelf PD-L1+ Human Natural Killer Cells Secreting IL15 to Treat Non-Small Cell Lung Cancer. Cancer immunology research. PMID 38572955
- **2025** Villalona-Calero MA et al. — Interim report on engineered NK cell trial in lung cancer refractory to immune checkpoint inhibitors. JCI insight. PMID 39903538 FT

Human translational

- **2026** Zhang TY et al. — Study on the Mechanisms of Allogeneic NK Cells Combined with Myeloid Cells in Treating Acute Myeloid Leukemia. Transplantation and cellular therapy. PMID 42035877

Process / manufacturing methods

- **2012** Della Chiesa M et al. — Phenotypic and functional heterogeneity of human NK cells developing after umbilical cord blood transplantation: a role for human cytomegalovirus?. *Blood*. PMID 22096237
- **2013** Ribeiro A et al. — Mesenchymal stem cells from umbilical cord matrix, adipose tissue and bone marrow exhibit different capability to suppress peripheral blood B, natural killer and T cells. *Stem cell research & therapy*. PMID 24406104 FT
- **2016** Koehl U et al. — Advances in clinical NK cell studies: Donor selection, manufacturing and quality control. *Oncoimmunology*. PMID 27141397 FT
- **2024** Huyghe M et al. — Comparative analysis of iPSC-derived NK cells from two differentiation strategies reveals distinct signatures and cytotoxic activities. *Frontiers in immunology*. PMID 39445004 FT

Preclinical (in vivo)

- **2013** Boissel L et al. — Retargeting NK-92 cells by means of CD19- and CD20-specific chimeric antigen receptors compares favorably with antibody-dependent cellular cytotoxicity. *Oncoimmunology*. PMID 24404423 FT
- **2020** Cichocki F et al. — iPSC-derived NK cells maintain high cytotoxicity and enhance in vivo tumor control in concert with T cells and anti-PD-1 therapy. *Science translational medicine*. PMID 33148626
- **2021** Myers JA et al. — Exploring the NK cell platform for cancer immunotherapy. *Nature reviews. Clinical oncology*. PMID 32934330
- **2024** Kong D et al. — CD19 CAR-expressing iPSC-derived NK cells effectively enhance migration and cytotoxicity into glioblastoma by targeting to the pericytes in tumor microenvironment. *Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie*. PMID 38508081

Mechanistic / in vitro

- **1994** Gong JH et al. — Characterization of a human cell line (NK-92) with phenotypical and functional characteristics of activated natural killer cells. *Leukemia*. PMID 8152260
- **2010** Cooley S et al. — Donor selection for natural killer cell receptor genes leads to superior survival after unrelated transplantation for acute myelogenous leukemia. *Blood*. PMID 20581313
- **2024** Thangaraj JL et al. — Disruption of TGF- β signaling pathway is required to mediate effective killing of hepatocellular carcinoma by human iPSC-derived NK cells. *Cell stem cell*. PMID 38986609

Reviews

- **2016** Klingemann H et al. — Natural Killer Cells for Immunotherapy - Advantages of the NK-92 Cell Line over Blood NK Cells. *Frontiers in immunology*. PMID 27014270 FT
- **2021** Heipertz EL et al. — Current Perspectives on "Off-The-Shelf" Allogeneic NK and CAR-NK Cell Therapies. *Frontiers in immunology*. PMID 34925314 FT
- **2023** Berrien-Elliott MM et al. — Allogeneic natural killer cell therapy. *Blood*. PMID 36416736 FT

CAR-NK & receptor engineering, gene delivery

32 papers · 2013–2025 · 22 with culture/manufacturing content

Clinical trials

- **2018** Murad JM et al. — Manufacturing development and clinical production of NKG2D chimeric antigen receptor-expressing T cells for autologous adoptive cell therapy. *Cytotherapy*. [PMID 30180944](#)
- **2019** Burger MC et al. — CAR-Engineered NK Cells for the Treatment of Glioblastoma: Turning Innate Effectors Into Precision Tools for Cancer Immunotherapy. *Frontiers in immunology*. [PMID 31798595](#) FT
- **2021** Sun M et al. — Novel two-chain structure utilizing KIRS2/DAP12 domain improves the safety and efficacy of CAR-T cells in adults with r/r B-ALL. *Molecular therapy oncolytics*. [PMID 34703879](#) FT

Human translational

- **2023** Powers SB et al. — Differential Expression of LLT1, SLAM Receptors CS1 and 2B4 and NCR Receptors NKp46 and NKp30 in Pediatric Acute Lymphoblastic Leukemia (ALL). *International journal of molecular sciences*. [PMID 36835271](#) FT

Process / manufacturing methods

- **2019** Ingegnere T et al. — Human CAR NK Cells: A New Non-viral Method Allowing High Efficient Transfection and Strong Tumor Cell Killing. *Frontiers in immunology*. [PMID 31114587](#) FT
- **2019** Müller S et al. — High Cytotoxic Efficiency of Lentivirally and Alpharetrovirally Engineered CD19-Specific Chimeric Antigen Receptor Natural Killer Cells Against Acute Lymphoblastic Leukemia. *Frontiers in immunology*. [PMID 32117200](#) FT
- **2020** Wang X et al. — Inducible MyD88/CD40 synergizes with IL-15 to enhance antitumor efficacy of CAR-NK cells. *Blood advances*. [PMID 32384544](#)

Preclinical (in vivo)

- **2013** Chang YH et al. — A chimeric receptor with NKG2D specificity enhances natural killer cell activation and killing of tumor cells. *Cancer research*. [PMID 23302231](#)
- **2014** Chu J et al. — CS1-specific chimeric antigen receptor (CAR)-engineered natural killer cells enhance in vitro and in vivo antitumor activity against human multiple myeloma. *Leukemia*. [PMID 24067492](#)
- **2015** Schönfeld K et al. — Selective inhibition of tumor growth by clonal NK cells expressing an ErbB2/HER2-specific chimeric antigen receptor. *Molecular therapy : the journal of the American Society of Gene Therapy*. [PMID 25373520](#)
- **2015** Han J et al. — CAR-Engineered NK Cells Targeting Wild-Type EGFR and EGFRvIII Enhance Killing of Glioblastoma and Patient-Derived Glioblastoma Stem Cells. *Scientific reports*. [PMID 26155832](#) FT
- **2018** Li Y et al. — Human iPSC-Derived Natural Killer Cells Engineered with Chimeric Antigen Receptors Enhance Anti-tumor Activity. *Cell stem cell*. [PMID 30082067](#)

- **2021** Daher M et al. — Targeting a cytokine checkpoint enhances the fitness of armored cord blood CAR-NK cells. *Blood*. PMID 32902645
- **2021** Christodoulou I et al. — Engineering CAR-NK cells to secrete IL-15 sustains their anti-AML functionality but is associated with systemic toxicities. *Journal for immunotherapy of cancer*. PMID 34896980 FT
- **2021** Ma R et al. — An Oncolytic Virus Expressing IL15/IL15R α Combined with Off-the-Shelf EGFR-CAR NK Cells Targets Glioblastoma. *Cancer research*. PMID 34006525
- **2022** Li Y et al. — KIR-based inhibitory CARs overcome CAR-NK cell trogocytosis-mediated fratricide and tumor escape. *Nature medicine*. PMID 36175679
- **2022** Da Y et al. — STING agonist cGAMP enhances anti-tumor activity of CAR-NK cells against pancreatic cancer. *Oncoimmunology*. PMID 35371622 FT
- **2023** Li L et al. — Loss of metabolic fitness drives tumor resistance after CAR-NK cell therapy and can be overcome by cytokine engineering. *Science advances*. PMID 37494448 FT
- **2024** Acharya S et al. — CD28 Costimulation Augments CAR Signaling in NK Cells via the LCK/CD3 ζ /ZAP70 Signaling Axis. *Cancer discovery*. PMID 38900051
- **2024** Albinger N et al. — Manufacturing of primary CAR-NK cells in an automated system for the treatment of acute myeloid leukemia. *Bone marrow transplantation*. PMID 38253870 FT
- **2025** Luo J et al. — Neoleukin-2/15-armored CAR-NK cells sustain superior therapeutic efficacy in solid tumors via c-Myc/NRF1 activation. *Signal transduction and targeted therapy*. PMID 40025022 FT
- **2025** Peng L et al. — In vivo AAV-SB-CRISPR screens of tumor-infiltrating primary NK cells identify genetic checkpoints of CAR-NK therapy. *Nature biotechnology*. PMID 38918616
- **2025** Yang S et al. — Non-pathogenic *E. coli* displaying decoy-resistant IL18 mutein boosts anti-tumor and CAR NK cell responses. *Nature biotechnology*. PMID 39367093

Mechanistic / in vitro

- **2022** Gurney M et al. — CD38 knockout natural killer cells expressing an affinity optimized CD38 chimeric antigen receptor successfully target acute myeloid leukemia with reduced effector cell fratricide. *Haematologica*. PMID 33375774 FT
- **2023** Dagher OK et al. — Forks in the road for CAR T and CAR NK cell cancer therapies. *Nature immunology*. PMID 38012406

Reviews

- **2018** Mehta RS et al. — Chimeric Antigen Receptor Expressing Natural Killer Cells for the Immunotherapy of Cancer. *Frontiers in immunology*. PMID 29497427 FT
- **2020** Xie G et al. — CAR-NK cells: A promising cellular immunotherapy for cancer. *EBioMedicine*. PMID 32853984 FT
- **2021** Gong Y et al. — Chimeric antigen receptor natural killer (CAR-NK) cell design and engineering for cancer therapy. *Journal of hematology & oncology*. PMID 33933160 FT
- **2022** Pan K et al. — CAR race to cancer immunotherapy: from CAR T, CAR NK to CAR macrophage therapy. *Journal of experimental & clinical cancer research : CR*. PMID 35361234 FT

- **2024** Wang W et al. — Breakthrough of solid tumor treatment: CAR-NK immunotherapy. Cell death discovery. [PMID 38245520](#) FT
- **2025** Balkhi S et al. — CAR-NK cell therapy: promise and challenges in solid tumors. Frontiers in immunology. [PMID 40260240](#) FT
- **2025** Jørgensen LV et al. — The clinical landscape of CAR NK cells. Experimental hematology & oncology. [PMID 40149002](#) FT

NK-cell engagers: BiKE, TriKE, NKp46/CD16 engagers

20 papers · 2013–2026 · 14 with culture/manufacturing content

Human translational

- **2021** Kerbaux LN et al. — Combining AFM13, a Bispecific CD30/CD16 Antibody, with Cytokine-Activated Blood and Cord Blood-Derived NK Cells Facilitates CAR-like Responses Against CD30+ Malignancies. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 33986022](#)
- **2021** Reusing SB et al. — CD16xCD33 Bispecific Killer Cell Engager (BiKE) as potential immunotherapeutic in pediatric patients with AML and biphenotypic ALL. Cancer immunology, immunotherapy : CII. [PMID 34398302](#) FT

Process / manufacturing methods

- **2016** Felices M et al. — Generation of BiKEs and TriKEs to Improve NK Cell-Mediated Targeting of Tumor Cells. Methods in molecular biology (Clifton, N.J.). [PMID 27177679](#)
- **2019** Felices M et al. — Novel CD19-targeted TriKE restores NK cell function and proliferative capacity in CLL. Blood advances. [PMID 30890546](#)
- **2025** Cheng Y et al. — An IL-15-modified NKp30×HER2 trispecific NK cell engager enhances NK cell activation and tumor cell killing. Journal of leukocyte biology. [PMID 40694640](#)

Preclinical (in vivo)

- **2016** Vallera DA et al. — IL15 Trispecific Killer Engagers (TriKE) Make Natural Killer Cells Specific to CD33+ Targets While Also Inducing Persistence, In Vivo Expansion, and Enhanced Function. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 26847056](#)
- **2021** Arvindam US et al. — A trispecific killer engager molecule against CLEC12A effectively induces NK-cell mediated killing of AML cells. Leukemia. [PMID 33097838](#) FT
- **2023** Gauthier L et al. — Control of acute myeloid leukemia by a trifunctional NKp46-CD16a-NK cell engager targeting CD123. Nature biotechnology. [PMID 36635380](#) FT
- **2023** Bordoloi D et al. — Siglec-7 glyco-immune binding mAbs or NK cell engager biologics induce potent antitumor immunity against ovarian cancers. Science advances. [PMID 37910620](#) FT
- **2024** Demaria O et al. — A tetraspecific engager armed with a non-alpha IL-2 variant harnesses natural killer cells against B cell non-Hodgkin lymphoma. Science immunology. [PMID 39546590](#)

- **2025** Khaw MJ et al. — Novel trisppecific killer engager targeting B7-H3 enhances natural killer cell antitumor activity against head and neck cancer. Journal for immunotherapy of cancer. [PMID 40707133](#) FT
- **2026** Fan Y et al. — Bispecific killer cell engager-secreting CAR-T cells redirect natural killer specificity to enhance antitumour responses. Nature biomedical engineering. [PMID 40659834](#)

Mechanistic / in vitro

- **2013** Wiernik A et al. — Targeting natural killer cells to acute myeloid leukemia in vitro with a CD16 x 33 bispecific killer cell engager and ADAM17 inhibition. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 23690482](#)
- **2014** Gleason MK et al. — CD16xCD33 bispecific killer cell engager (BiKE) activates NK cells against primary MDS and MDSC CD33+ targets. Blood. [PMID 24652987](#)
- **2015** Wu J et al. — AFM13: a first-in-class tetravalent bispecific anti-CD30/CD16A antibody for NK cell-mediated immunotherapy. Journal of hematology & oncology. [PMID 26231785](#) FT
- **2022** Nikkhoi SK et al. — Bispecific killer cell engager with high affinity and specificity toward CD16a on NK cells for cancer immunotherapy. Frontiers in immunology. [PMID 36685519](#) FT
- **2025** Zambarda C et al. — CD16A Shedding Regulates Innate Cell Engager-Induced Serial Killing by Natural Killer Cells. European journal of immunology. [PMID 41116262](#) FT

Reviews

- **2019** Hodgins JJ et al. — Killers 2.0: NK cell therapies at the forefront of cancer control. The Journal of clinical investigation. [PMID 31478911](#)
- **2023** Zhang M et al. — Natural Killer Cell Engagers (NKCEs): a new frontier in cancer immunotherapy. Frontiers in immunology. [PMID 37638058](#) FT
- **2024** Shin MH et al. — Current Developments in NK Cell Engagers for Cancer Immunotherapy: Focus on CD16A and NKp46. Immune network. [PMID 39513028](#) FT

Gene editing: checkpoint and brake knockouts

24 papers · 2014–2025 · 18 with culture/manufacturing content

Human translational

- **2023** Bi J et al. — TIPE2 deletion improves the therapeutic potential of adoptively transferred NK cells. Journal for immunotherapy of cancer. [PMID 36725083](#) FT
- **2024** Xu Y et al. — Enhanced NK cell activation via eEF2K-mediated potentiation of the cGAS-STING pathway in hepatocellular carcinoma. International immunopharmacology. [PMID 38320351](#)

Process / manufacturing methods

- **2021** Huang RS et al. — A robust platform for expansion and genome editing of primary human natural killer cells. The Journal of experimental medicine. [PMID 33433623](#) FT
- **2021** Morimoto T et al. — CRISPR-Cas9-Mediated TIM3 Knockout in Human Natural Killer Cells Enhances Growth Inhibitory Effects on Human Glioma Cells. International journal of molecular sciences. [PMID 33800561](#) FT
- **2022** Bexte T et al. — CRISPR-Cas9 based gene editing of the immune checkpoint NKG2A enhances NK cell mediated cytotoxicity against multiple myeloma. Oncoimmunology. [PMID 35694192](#) FT

Preclinical (in vivo)

- **2014** Paolino M et al. — The E3 ligase Cbl-b and TAM receptors regulate cancer metastasis via natural killer cells. Nature. [PMID 24553136](#)
- **2017** Tang PM et al. — Smad3 promotes cancer progression by inhibiting E4BP4-mediated NK cell development. Nature communications. [PMID 28262747](#) FT
- **2020** Zhu H et al. — Metabolic Reprograming via Deletion of CISH in Human iPSC-Derived NK Cells Promotes In Vivo Persistence and Enhances Anti-tumor Activity. Cell stem cell. [PMID 32531207](#)
- **2020** Naeimi Kararoudi M et al. — CD38 deletion of human primary NK cells eliminates daratumumab-induced fratricide and boosts their effector activity. Blood. [PMID 32603414](#)
- **2020** Huang RS et al. — Enhanced NK-92 Cytotoxicity by CRISPR Genome Engineering Using Cas9 Ribonucleoproteins. Frontiers in immunology. [PMID 32528479](#) FT
- **2020** Hoerster K et al. — HLA Class I Knockout Converts Allogeneic Primary NK Cells Into Suitable Effectors for "Off-the-Shelf" Immunotherapy. Frontiers in immunology. [PMID 33584651](#) FT
- **2022** Bernareggi D et al. — CHMP2A regulates tumor sensitivity to natural killer cell-mediated cytotoxicity. Nature communications. [PMID 35393416](#) FT
- **2022** Bernard PL et al. — Targeting CISH enhances natural cytotoxicity receptor signaling and reduces NK cell exhaustion to improve solid tumor immunity. Journal for immunotherapy of cancer. [PMID 35589278](#) FT
- **2022** Clara JA et al. — High-affinity CD16 integration into a CRISPR/Cas9-edited CD38 locus augments CD38-directed antitumor activity of primary human natural killer cells. Journal for immunotherapy of cancer. [PMID 35135865](#) FT
- **2023** Mac Donald A et al. — KLRC1 knockout overcomes HLA-E-mediated inhibition and improves NK cell antitumor activity against solid tumors. Frontiers in immunology. [PMID 37675109](#) FT
- **2024** Bexte T et al. — CRISPR/Cas9 editing of NKG2A improves the efficacy of primary CD33-directed chimeric antigen receptor natural killer cells. Nature communications. [PMID 39349459](#) FT
- **2024** Reggiani F et al. — BET inhibitors drive Natural Killer activation in non-small cell lung cancer via BRD4 and SMAD3. Nature communications. [PMID 38519469](#) FT
- **2024** Sun X et al. — Deletion of the mRNA endonuclease Regnase-1 promotes NK cell anti-tumor activity via OCT2-dependent transcription of Ifng. Immunity. [PMID 38821052](#)

- **2024** Matson AW et al. — Enhanced IL-15-mediated NK cell activation and proliferation by an ADAM17 function-blocking antibody involves CD16A, CD137, and accessory cells. Journal for immunotherapy of cancer. [PMID 39053944](#) FT
- **2025** Biederstädt A et al. — Genome-wide CRISPR screens identify critical targets to enhance CAR-NK cell antitumor potency. Cancer cell. [PMID 40845844](#)

Mechanistic / in vitro

- **2020** Pomeroy EJ et al. — A Genetically Engineered Primary Human Natural Killer Cell Platform for Cancer Immunotherapy. Molecular therapy : the journal of the American Society of Gene Therapy. [PMID 31704085](#)
- **2025** Rea A et al. — Enhancing human NK cell antitumor function by knocking out SMAD4 to counteract TGFβ and activin A suppression. Nature immunology. [PMID 40119192](#) FT

Reviews

- **2019** Afolabi LO et al. — Genetic reprogramming for NK cell cancer immunotherapy with CRISPR/Cas9. Immunology. [PMID 31315144](#)
- **2025** Fazeli A et al. — Engineering with care: safety assessment platforms for CRISPR-modified natural killer cells. Frontiers in immunology. [PMID 41425600](#) FT

Cytokine support & armouring (IL-15, IL-21, IL-12/15/18, IL-18)

24 papers · 1993–2025 · 18 with culture/manufacturing content

Clinical trials

- **2009** Viaud S et al. — Dendritic cell-derived exosomes promote natural killer cell activation and proliferation: a role for NKG2D ligands and IL-15Ralpha. PloS one. [PMID 19319200](#) FT
- **2015** Shah NN et al. — Acute GVHD in patients receiving IL-15/4-1BBL activated NK cells following T-cell-depleted stem cell transplantation. Blood. [PMID 25452614](#)
- **2016** Romee R et al. — Cytokine-induced memory-like natural killer cells exhibit enhanced responses against myeloid leukemia. Science translational medicine. [PMID 27655849](#)
- **2022** Berrien-Elliott MM et al. — Systemic IL-15 promotes allogeneic cell rejection in patients treated with natural killer cell adoptive therapy. Blood. [PMID 34797911](#)

Human translational

- **2018** Al-Attar A et al. — Human Body Composition and Immunity: Visceral Adipose Tissue Produces IL-15 and Muscle Strength Inversely Correlates with NK Cell Function in Elderly Humans. Frontiers in immunology. [PMID 29559978](#) FT
- **2024** Li Y et al. — DNMT1 inhibition improves the activity of memory-like natural killer cells by enhancing the level of autophagy. Molecular biology reports. [PMID 39704855](#) FT

Process / manufacturing methods

- **2012** Sahm C et al. — Expression of IL-15 in NK cells results in rapid enrichment and selective cytotoxicity of gene-modified effectors that carry a tumor-specific antigen receptor. *Cancer immunology, immunotherapy* : CII. [PMID 22310931](#)
- **2019** Choi YH et al. — IL-27 enhances IL-15/IL-18-mediated activation of human natural killer cells. *Journal for immunotherapy of cancer*. [PMID 31277710](#) FT
- **2019** Kweon S et al. — Expansion of Human NK Cells Using K562 Cells Expressing OX40 Ligand and Short Exposure to IL-21. *Frontiers in immunology*. [PMID 31105701](#) FT

Preclinical (in vivo)

- **1993** D'Andrea A et al. — Interleukin 10 (IL-10) inhibits human lymphocyte interferon gamma-production by suppressing natural killer cell stimulatory factor/IL-12 synthesis in accessory cells. *The Journal of experimental medicine*. [PMID 8102388](#) FT
- **1999** Kitamura H et al. — The natural killer T (NKT) cell ligand alpha-galactosylceramide demonstrates its immunopotentiating effect by inducing interleukin (IL)-12 production by dendritic cells and IL-12 receptor expression on NKT cells. *The Journal of experimental medicine*. [PMID 10190903](#) FT
- **2009** Huntington ND et al. — IL-15 trans-presentation promotes human NK cell development and differentiation in vivo. *The Journal of experimental medicine*. [PMID 19103877](#) FT
- **2012** Denman CJ et al. — Membrane-bound IL-21 promotes sustained ex vivo proliferation of human natural killer cells. *PloS one*. [PMID 22279576](#) FT
- **2012** Ni J et al. — Sustained effector function of IL-12/15/18-preactivated NK cells against established tumors. *The Journal of experimental medicine*. [PMID 23209317](#) FT
- **2018** Liu E et al. — Cord blood NK cells engineered to express IL-15 and a CD19-targeted CAR show long-term persistence and potent antitumor activity. *Leukemia*. [PMID 28725044](#)
- **2024** Van den Eynde A et al. — IL-15-secreting CAR natural killer cells directed toward the pan-cancer target CD70 eliminate both cancer cells and cancer-associated fibroblasts. *Journal of hematology & oncology*. [PMID 38331849](#) FT
- **2024** Shanley M et al. — Interleukin-21 engineering enhances NK cell activity against glioblastoma via CEBPD. *Cancer cell*. [PMID 39137729](#)
- **2024** Berjis A et al. — Pretreatment with IL-15 and IL-18 rescues natural killer cells from granzyme B-mediated apoptosis after cryopreservation. *Nature communications*. [PMID 38729924](#) FT
- **2025** Rafei H et al. — CREM is a regulatory checkpoint of CAR and IL-15 signalling in NK cells. *Nature*. [PMID 40468083](#) FT

Mechanistic / in vitro

- **1994** Carson WE et al. — Interleukin (IL) 15 is a novel cytokine that activates human natural killer cells via components of the IL-2 receptor. *The Journal of experimental medicine*. [PMID 7523571](#) FT
- **2012** Ndhlovu LC et al. — Tim-3 marks human natural killer cell maturation and suppresses cell-mediated cytotoxicity. *Blood*. [PMID 22383801](#)

- **2024** Kennedy PR et al. — Metabolic programs drive function of therapeutic NK cells in hypoxic tumor environments. Science advances. [PMID 39475618](#) FT

Reviews

- **2015** Carlsten M et al. — Genetic Manipulation of NK Cells for Cancer Immunotherapy: Techniques and Clinical Implications. Frontiers in immunology. [PMID 26113846](#) FT
- **2018** Abel AM et al. — Natural Killer Cells: Development, Maturation, and Clinical Utilization. Frontiers in immunology. [PMID 30150991](#) FT

Ex vivo expansion, feeders, media & GMP manufacturing

30 papers · 2009–2025 · 30 with culture/manufacturing content

Clinical trials

- **2010** Cho D et al. — Cytotoxicity of activated natural killer cells against pediatric solid tumors. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 20542985](#)
- **2014** Rölle A et al. — IL-12-producing monocytes and HLA-E control HCMV-driven NKG2C+ NK cell expansion. The Journal of clinical investigation. [PMID 25384219](#)
- **2016** Leivas A et al. — Novel treatment strategy with autologous activated and expanded natural killer cells plus anti-myeloma drugs for multiple myeloma. Oncoimmunology. [PMID 28123890](#) FT

Human translational

- **2019** Zhu H et al. — An Improved Method to Produce Clinical-Scale Natural Killer Cells from Human Pluripotent Stem Cells. Methods in molecular biology (Clifton, N.J.). [PMID 31396935](#)
- **2023** Mou J et al. — 2B4 inhibits the apoptosis of natural killer cells through phosphorylated extracellular signal-related kinase/B-cell lymphoma 2 signal pathway. Cytotherapy. [PMID 37516949](#)

Process / manufacturing methods

- **2009** Cho D et al. — Expansion and activation of natural killer cells for cancer immunotherapy. The Korean journal of laboratory medicine. [PMID 19411773](#)
- **2010** Sutlu T et al. — Clinical-grade, large-scale, feeder-free expansion of highly active human natural killer cells for adoptive immunotherapy using an automated bioreactor. Cytotherapy. [PMID 20795758](#)
- **2012** Lapteva N et al. — Large-scale ex vivo expansion and characterization of natural killer cells for clinical applications. Cytotherapy. [PMID 22900959](#)
- **2013** Knorr DA et al. — Clinical-scale derivation of natural killer cells from human pluripotent stem cells for cancer therapy. Stem cells translational medicine. [PMID 23515118](#)
- **2013** Kang L et al. — Characterization and ex vivo Expansion of Human Placenta-Derived Natural Killer Cells for Cancer Immunotherapy. Frontiers in immunology. [PMID 23641243](#) FT

- **2015** Huijskens MJ et al. — Ascorbic acid promotes proliferation of natural killer cell populations in culture systems applicable for natural killer cell therapy. *Cytotherapy*. PMID 25747742
- **2017** Klöß S et al. — Optimization of Human NK Cell Manufacturing: Fully Automated Separation, Improved Ex Vivo Expansion Using IL-21 with Autologous Feeder Cells, and Generation of Anti-CD123-CAR-Expressing Effector Cells. *Human gene therapy*. PMID 28810809
- **2019** Fernández L et al. — GMP-Compliant Manufacturing of NKG2D CAR Memory T Cells Using CliniMACS Prodigy. *Frontiers in immunology*. PMID 31649672 FT
- **2020** Reina-Ortiz C et al. — Expanded NK cells from umbilical cord blood and adult peripheral blood combined with daratumumab are effective against tumor cells from multiple myeloma patients. *Oncoimmunology*. PMID 33457074 FT
- **2024** Wang X et al. — Scalable process development of NK and CAR-NK expansion in a closed bioreactor. *Frontiers in immunology*. PMID 39114666 FT
- **2025** Tran TBT et al. — In Vitro Expansion and Transduction of Primary NK Cells Using Feeder Cells Expressing Costimulatory Molecules and IL-21. *Cancer science*. PMID 40325497 FT
- **2025** Kikuchi T et al. — Scalable production process development for NK cells targeting large-scale expansion. *Regenerative therapy*. PMID 40808775 FT

Preclinical (in vivo)

- **2009** Fujisaki H et al. — Expansion of highly cytotoxic human natural killer cells for cancer cell therapy. *Cancer research*. PMID 19383914
- **2010** Siegler U et al. — Good manufacturing practice-compliant cell sorting and large-scale expansion of single KIR-positive alloreactive human natural killer cells for multiple infusions to leukemia patients. *Cytotherapy*. PMID 20491532
- **2011** Somanchi SS et al. — Expansion, purification, and functional assessment of human peripheral blood NK cells. *Journal of visualized experiments : JoVE*. PMID 21339714
- **2013** Shah N et al. — Antigen presenting cell-mediated expansion of human umbilical cord blood yields log-scale expansion of natural killer cells with anti-myeloma activity. *PloS one*. PMID 24204673 FT
- **2016** Granzin M et al. — Highly efficient IL-21 and feeder cell-driven ex vivo expansion of human NK cells with therapeutic activity in a xenograft mouse model of melanoma. *Oncoimmunology*. PMID 27757317
- **2016** Kamiya T et al. — Expanded and Activated Natural Killer Cells for Immunotherapy of Hepatocellular Carcinoma. *Cancer immunology research*. PMID 27197065
- **2016** Oyer JL et al. — Natural killer cells stimulated with PM21 particles expand and biodistribute in vivo: Clinical implications for cancer treatment. *Cytotherapy*. PMID 27059202
- **2019** Ojo EO et al. — Membrane bound IL-21 based NK cell feeder cells drive robust expansion and metabolic activation of NK cells. *Scientific reports*. PMID 31624330 FT
- **2023** Sun Y et al. — Macrophage STING signaling promotes NK cell to suppress colorectal cancer liver metastasis via 4-1BBL/4-1BB co-stimulation. *Journal for immunotherapy of cancer*. PMID 36927529 FT

- **2024** Oyer JL et al. — PM21-particle stimulation augmented with cytokines enhances NK cell expansion and confers memory-like characteristics with enhanced survival. *Frontiers in immunology*. [PMID 38711506](#) FT
- **2025** Lin H et al. — Optimizing the procedure for manufacturing clinical-grade genetically manipulated natural killer cells for adoptive immunotherapy. *Cytotherapy*. [PMID 39570247](#)

Reviews

- **2022** Gurney M et al. — Feeder Cells at the Interface of Natural Killer Cell Activation, Expansion and Gene Editing. *Frontiers in immunology*. [PMID 35222382](#) FT
- **2025** Lee J et al. — Recent advances in feeder-free NK cell expansion as a future direction of adoptive cell therapy. *Frontiers in immunology*. [PMID 41089682](#) FT

Cryopreservation, potency assays & product quality

15 papers · 2015–2026 · 15 with culture/manufacturing content

Process / manufacturing methods

- **2015** Vasu S et al. — A novel method to expand large numbers of CD56(+) natural killer cells from a minute fraction of selectively accessed cryopreserved cord blood for immunotherapy after transplantation. *Cytotherapy*. [PMID 26432560](#)
- **2018** Nham T et al. — Ex Vivo-expanded Natural Killer Cells Derived From Long-term Cryopreserved Cord Blood are Cytotoxic Against Primary Breast Cancer Cells. *Journal of immunotherapy (Hagerstown, Md. : 1997)*. [PMID 29189387](#)
- **2020** Yao X et al. — Nanoparticle-Mediated Intracellular Protection of Natural Killer Cells Avoids Cryoinjury and Retains Potent Antitumor Functions. *Advanced science (Weinheim, Baden-Wurttemberg, Germany)*. [PMID 32382476](#) FT
- **2024** Lee S et al. — Successful expansion and cryopreservation of human natural killer cell line NK-92 for clinical manufacturing. *PloS one*. [PMID 38394177](#) FT
- **2025** Joules A et al. — Probing mechanisms of cryopreservation damage to natural killer cells. *Cytotherapy*. [PMID 39918490](#)
- **2025** Zhou X et al. — Restoring NK Cell Cytotoxicity Post-Cryopreservation via Synthetic Cells. *Advanced science (Weinheim, Baden-Wurttemberg, Germany)*. [PMID 40966444](#) FT
- **2025** Kikuchi T et al. — Evaluation of several type of chemical cryoprotectants and combination towards cryopreservation of NK cell. *Cytotechnology*. [PMID 41208818](#)
- **2026** Joules A et al. — Natural killer cell cryopreservation: Influence of composition, temperature, and granules on post-thaw survival and function. *Molecular therapy. Advances*. [PMID 42564981](#) FT
- **2026** Liu Y et al. — Induction of stress granules alleviates programmed cell death induced by lysosomal damage during NK cell cryopreservation. *Cell death discovery*. [PMID 42098076](#) FT

Mechanistic / in vitro

- **2016** Holubova M et al. — Cryopreserved NK cells in the treatment of haematological malignancies: preclinical study. Journal of cancer research and clinical oncology. PMID 27614454
- **2019** Mark C et al. — Cryopreservation impairs cytotoxicity and migration of NK cells in 3-D tissue: Implications for cancer immunotherapy. bioRxiv. 10.1101/812172
- **2020** Mark C et al. — Cryopreservation impairs 3-D migration and cytotoxicity of natural killer cells. Nature communications. PMID 33067467 FT
- **2020** Damodharan SN et al. — Analysis of ex vivo expanded and activated clinical-grade human NK cells after cryopreservation. Cytotherapy. PMID 32536506
- **2022** Reusch U et al. — Cryopreservation of Natural Killer Cells Pre-Complexed with Innate Cell Engagers. Antibodies (Basel, Switzerland). PMID 35225870 FT
- **2024** St-Denis-Bissonnette F et al. — Evaluation of resazurin phenoxazine dye as a highly sensitive cell viability potency assay for natural killer cell-derived extracellular vesicle-based cancer biotherapeutics. Journal of extracellular biology. PMID 39022723 FT

Metabolic fitness, hypoxia & in vivo persistence

18 papers · 2012–2025 · 11 with culture/manufacturing content

Human translational

- **2021** Slattery K et al. — TGFβ drives NK cell metabolic dysfunction in human metastatic breast cancer. Journal for immunotherapy of cancer. PMID 33568351 FT
- **2024** Portale F et al. — C/EBPβ-dependent autophagy inhibition hinders NK cell function in cancer. Nature communications. PMID 39609420 FT
- **2025** von Kries A et al. — Colorectal cancer organoids drive hypoxia, TGF-β, and patient-specific diversification of NK cell activation programs. Journal for immunotherapy of cancer. PMID 41360426 FT

Process / manufacturing methods

- **2013** Sarkar S et al. — Hypoxia induced impairment of NK cell cytotoxicity against multiple myeloma can be overcome by IL-2 activation of the NK cells. PloS one. PMID 23724099 FT
- **2018** Xu LJ et al. — Combined inhibition of JAK1,2/Stat3-PD-L1 signaling pathway suppresses the immune escape of castration-resistant prostate cancer to NK cells in hypoxia. Molecular medicine reports. PMID 29693186 FT
- **2020** Teng R et al. — Hypoxia Impairs NK Cell Cytotoxicity through SHP-1-Mediated Attenuation of STAT3 and ERK Signaling Pathways. Journal of immunology research. PMID 33123602 FT

Preclinical (in vivo)

- **2012** Sceneay J et al. — Primary tumor hypoxia recruits CD11b+/Ly6Cmed/Ly6G+ immune suppressor cells and compromises NK cell cytotoxicity in the premetastatic niche. Cancer research. PMID 22751463

- **2013** Baginska J et al. — Granzyme B degradation by autophagy decreases tumor cell susceptibility to natural killer-mediated lysis under hypoxia. *Proceedings of the National Academy of Sciences of the United States of America*. PMID 24101526
- **2016** Viel S et al. — TGF- β inhibits the activation and functions of NK cells by repressing the mTOR pathway. *Science signaling*. PMID 26884601
- **2018** Michelet X et al. — Metabolic reprogramming of natural killer cells in obesity limits antitumor responses. *Nature immunology*. PMID 30420624
- **2018** Loftus RM et al. — Amino acid-dependent cMyc expression is essential for NK cell metabolic and functional responses in mice. *Nature communications*. PMID 29904050 FT
- **2022** Pan R et al. — Augmenting NK cell-based immunotherapy by targeting mitochondrial apoptosis. *Cell*. PMID 35447071
- **2024** Wu S et al. — The ω -3 Polyunsaturated Fatty Acid Docosahexaenoic Acid Enhances NK-Cell Antitumor Effector Functions. *Cancer immunology research*. PMID 38526128 FT

Mechanistic / in vitro

- **2024** Meng M et al. — mTOR Signaling Promotes Rapid m6A mRNA Methylation to Regulate NK-Cell Activation and Effector Functions. *Cancer immunology research*. PMID 38640466
- **2025** Wei C et al. — Nuclear mitochondrial acetyl-CoA acetyltransferase 1 orchestrates natural killer cell-dependent antitumor immunity in colorectal cancer. *Signal transduction and targeted therapy*. PMID 40289129 FT

Reviews

- **2019** Terrén I et al. — NK Cell Metabolism and Tumor Microenvironment. *Frontiers in immunology*. PMID 31616440 FT
- **2024** Miao L et al. — Advances in metabolic reprogramming of NK cells in the tumor microenvironment on the impact of NK therapy. *Journal of translational medicine*. PMID 38433193 FT
- **2025** Ghosh S et al. — Mitochondrial dynamics and metabolic attributes regulate function of natural killer cell and infiltration in tumor microenvironment modulating disease progression. *Biochimica et biophysica acta. Reviews on cancer*. PMID 41075850

Inhibitory receptors, checkpoints & tumour escape

18 papers · 2010–2025 · 13 with culture/manufacturing content

Clinical trials

- **2010** Benson DM et al. — The PD-1/PD-L1 axis modulates the natural killer cell versus multiple myeloma effect: a therapeutic target for CT-011, a novel monoclonal anti-PD-1 antibody. *Blood*. PMID 20460501
- **2018** André P et al. — Anti-NKG2A mAb Is a Checkpoint Inhibitor that Promotes Anti-tumor Immunity by Unleashing Both T and NK Cells. *Cell*. PMID 30503213 FT
- **2018** Vari F et al. — Immune evasion via PD-1/PD-L1 on NK cells and monocyte/macrophages is more prominent in Hodgkin lymphoma than DLBCL. *Blood*. PMID 29449276

- **2024** Xiong J et al. — Anti-metabolic agent pegaspargase plus PD-1 antibody sintilimab for first-line treatment in advanced natural killer T cell lymphoma. Signal transduction and targeted therapy. [PMID 38448403](#) FT

Human translational

- **2025** Sun Y et al. — Platelet-mediated circulating tumor cell evasion from natural killer cell killing through immune checkpoint CD155-TIGIT. Hepatology (Baltimore, Md.). [PMID 38779918](#)
- **2025** Wang J et al. — Differential impact of TIM-3 ligands on NK cell function. Journal for immunotherapy of cancer. [PMID 39773563](#) FT

Process / manufacturing methods

- **2018** Lanuza PM et al. — Activated human primary NK cells efficiently kill colorectal cancer cells in 3D spheroid cultures irrespectively of the level of PD-L1 expression. Oncoimmunology. [PMID 29632716](#)
- **2018** Xu L et al. — Inhibition of IL-6-JAK/Stat3 signaling in castration-resistant prostate cancer cells enhances the NK cell-mediated cytotoxicity via alteration of PD-L1/NKG2D ligand levels. Molecular oncology. [PMID 28865178](#) FT
- **2019** Carlsten M et al. — Bortezomib sensitizes multiple myeloma to NK cells via ER-stress-induced suppression of HLA-E and upregulation of DR5. Oncoimmunology. [PMID 30713790](#) FT

Preclinical (in vivo)

- **2018** Zhang Q et al. — Blockade of the checkpoint receptor TIGIT prevents NK cell exhaustion and elicits potent anti-tumor immunity. Nature immunology. [PMID 29915296](#)
- **2018** Hsu J et al. — Contribution of NK cells to immunotherapy mediated by PD-1/PD-L1 blockade. The Journal of clinical investigation. [PMID 30198904](#)
- **2023** Liu X et al. — Immune checkpoint HLA-E:CD94-NKG2A mediates evasion of circulating tumor cells from NK cell surveillance. Cancer cell. [PMID 36706761](#)
- **2024** Taylor BC et al. — NKG2A Is a Therapeutic Vulnerability in Immunotherapy Resistant MHC-I Heterogeneous Triple-Negative Breast Cancer. Cancer discovery. [PMID 37791898](#)
- **2025** Xie J et al. — Overexpressing natural killer group 2 member A drives natural killer cell exhaustion in relapsed acute myeloid leukemia. Signal transduction and targeted therapy. [PMID 40320412](#) FT

Mechanistic / in vitro

- **2024** Lupo KB et al. — synNotch-programmed iPSC-derived NK cells usurp TIGIT and CD73 activities for glioblastoma therapy. Nature communications. [PMID 38429294](#) FT

Reviews

- **2019** Pende D et al. — Killer Ig-Like Receptors (KIRs): Their Role in NK Cell Modulation and Developments Leading to Their Clinical Exploitation. Frontiers in immunology. [PMID 31231370](#) FT

- **2020** Khan M et al. — NK Cell-Based Immune Checkpoint Inhibition. *Frontiers in immunology*. [PMID 32117298](#) FT
- **2024** Li Y et al. — Unlocking the therapeutic potential of the NKG2A-HLA-E immune checkpoint pathway in T cells and NK cells for cancer immunotherapy. *Journal for immunotherapy of cancer*. [PMID 39486805](#) FT

ADCC, antibody and drug combinations

18 papers · 2008–2025 · 15 with culture/manufacturing content

Clinical trials

- **2017** Canter RJ et al. — Radiotherapy enhances natural killer cell cytotoxicity and localization in pre-clinical canine sarcomas and first-in-dog clinical trial. *Journal for immunotherapy of cancer*. [PMID 29254507](#) FT
- **2022** Walle T et al. — Radiotherapy orchestrates natural killer cell dependent antitumor immune responses through CXCL8. *Science advances*. [PMID 35319989](#) FT
- **2022** Nakamura A et al. — The SUMOylation inhibitor subasumstat potentiates rituximab activity by IFN1-dependent macrophage and NK cell stimulation. *Blood*. [PMID 35226739](#) FT
- **2024** Choi MG et al. — Safety and efficacy of SNK01 (autologous natural killer cells) in combination with cytotoxic chemotherapy and/or cetuximab after failure of prior tyrosine kinase inhibitor in non-small cell lung cancer: non-clinical mouse model and phase I/IIa clinical study. *Journal for immunotherapy of cancer*. [PMID 38538093](#) FT

Human translational

- **2015** Jie HB et al. — CTLA-4⁺ Regulatory T Cells Increased in Cetuximab-Treated Head and Neck Cancer Patients Suppress NK Cell Cytotoxicity and Correlate with Poor Prognosis. *Cancer research*. [PMID 25832655](#)
- **2020** Park JE et al. — Anti-tumor effects of NK cells and anti-PD-L1 antibody with antibody-dependent cellular cytotoxicity in PD-L1-positive cancer cell lines. *Journal for immunotherapy of cancer*. [PMID 32830112](#) FT
- **2024** Kumar B et al. — BATF is a major driver of NK cell epigenetic reprogramming and dysfunction in AML. *Science translational medicine*. [PMID 39259809](#)

Process / manufacturing methods

- **2008** Binyamin L et al. — Blocking NK cell inhibitory self-recognition promotes antibody-dependent cellular cytotoxicity in a model of anti-lymphoma therapy. *Journal of immunology* (Baltimore, Md. : 1950). [PMID 18424763](#)
- **2021** Gopal S et al. — 3D tumor spheroid microarray for high-throughput, high-content natural killer cell-mediated cytotoxicity. *Communications biology*. [PMID 34290356](#) FT
- **2025** Maskalenko NA et al. — The FcγRIIIA (CD16) L48-H/R Polymorphism Enhances NK Cell-Mediated Antibody-Dependent Cellular Cytotoxicity by Promoting Serial Killing. *Cancer immunology research*. [PMID 39666369](#)

Preclinical (in vivo)

- **2015** Yin J et al. — Ezh2 regulates differentiation and function of natural killer cells through histone methyltransferase activity. Proceedings of the National Academy of Sciences of the United States of America. [PMID 26668377](#)
- **2018** Weiss T et al. — NKG2D-Based CAR T Cells and Radiotherapy Exert Synergistic Efficacy in Glioblastoma. Cancer research. [PMID 29222400](#)
- **2020** Zhu H et al. — Pluripotent stem cell-derived NK cells with high-affinity noncleavable CD16a mediate improved antitumor activity. Blood. [PMID 31856277](#)
- **2024** Arora J et al. — T-cell help in the tumor microenvironment enhances rituximab-mediated NK-cell ADCC. Blood. [PMID 38457360](#)

Mechanistic / in vitro

- **2024** Liu J et al. — Epigenetic regulation of CD38/CD48 by KDM6A mediates NK cell response in multiple myeloma. Nature communications. [PMID 38355622](#) FT

Reviews

- **2013** Seidel UJ et al. — Natural killer cell mediated antibody-dependent cellular cytotoxicity in tumor immunotherapy with therapeutic antibodies. Frontiers in immunology. [PMID 23543707](#) FT
- **2015** Wang W et al. — NK Cell-Mediated Antibody-Dependent Cellular Cytotoxicity in Cancer Immunotherapy. Frontiers in immunology. [PMID 26284063](#) FT
- **2025** Vincken R et al. — ADCC: the rock band led by therapeutic antibodies, tumor and immune cells. Frontiers in immunology. [PMID 40308580](#) FT

Trafficking, homing & the solid-tumour microenvironment

18 papers · 2003–2025 · 12 with culture/manufacturing content

Clinical trials

- **2012** Somanchi SS et al. — Engineering lymph node homing of ex vivo-expanded human natural killer cells via trogocytosis of the chemokine receptor CCR7. Blood. [PMID 22498742](#)
- **2020** Okoye I et al. — Galectin-9 expression defines exhausted T cells and impaired cytotoxic NK cells in patients with virus-associated solid tumors. Journal for immunotherapy of cancer. [PMID 33310773](#) FT
- **2022** Schomer NT et al. — CCR7 expression in CD19 chimeric antigen receptor-engineered natural killer cells improves migration toward CCL19-expressing lymphoma cells and increases tumor control in mice with human lymphoma. Cytotherapy. [PMID 35400595](#)

Human translational

- **2024** Neo SY et al. — Tumor-associated NK cells drive MDSC-mediated tumor immune tolerance through the IL-6/STAT3 axis. Science translational medicine. [PMID 38748775](#)

- **2024** Fang H et al. — myCAF-derived exosomal PWAR6 accelerates CRC liver metastasis via altering glutamine availability and NK cell function in the tumor microenvironment. *Journal of hematology & oncology*. [PMID 39696364](#) FT
- **2025** Russo E et al. — Cxcr3 promotes protection from colorectal cancer liver metastasis by driving NK cell infiltration and plasticity. *The Journal of clinical investigation*. [PMID 40168086](#) FT

Process / manufacturing methods

- **2018** Huergo-Zapico L et al. — NK-cell Editing Mediates Epithelial-to-Mesenchymal Transition via Phenotypic and Proteomic Changes in Melanoma Cell Lines. *Cancer research*. [PMID 29752261](#)
- **2022** Marzagalli M et al. — A multi-organ-on-chip to recapitulate the infiltration and the cytotoxic activity of circulating NK cells in 3D matrix-based tumor model. *Frontiers in bioengineering and biotechnology*. [PMID 35957642](#) FT
- **2024** Karlsson V et al. — Elevated Galectin-3 levels in the tumor microenvironment of ovarian cancer - implication of ROS mediated suppression of NK cell antitumor response via tumor-associated neutrophils. *Frontiers in immunology*. [PMID 39759523](#) FT

Preclinical (in vivo)

- **2003** Ancuta P et al. — Fractalkine preferentially mediates arrest and migration of CD16+ monocytes. *The Journal of experimental medicine*. [PMID 12810688](#) FT
- **2018** Böttcher JP et al. — NK Cells Stimulate Recruitment of cDC1 into the Tumor Microenvironment Promoting Cancer Immune Control. *Cell*. [PMID 29429633](#) FT
- **2018** Barry KC et al. — A natural killer-dendritic cell axis defines checkpoint therapy-responsive tumor microenvironments. *Nature medicine*. [PMID 29942093](#)
- **2024** He C et al. — Vitamin B6 Competition in the Tumor Microenvironment Hampers Antitumor Functions of NK Cells. *Cancer discovery*. [PMID 37931287](#) FT
- **2025** Ben-Shmuel A et al. — Cancer-Associated Fibroblasts Serve as Decoys to Suppress NK Cell Anticancer Cytotoxicity in Breast Cancer. *Cancer discovery*. [PMID 40052789](#)
- **2025** Jin J et al. — Targeting lactylation reinforces NK cell cytotoxicity within the tumor microenvironment. *Nature immunology*. [PMID 40494934](#)

Reviews

- **2020** Shimasaki N et al. — NK cells for cancer immunotherapy. *Nature reviews. Drug discovery*. [PMID 31907401](#)
- **2022** Ran GH et al. — Natural killer cell homing and trafficking in tissues and tumors: from biology to application. *Signal transduction and targeted therapy*. [PMID 35768424](#) FT
- **2024** Peng L et al. — CAR-T and CAR-NK as cellular cancer immunotherapy for solid tumors. *Cellular & molecular immunology*. [PMID 39134804](#) FT

Clinical evidence: adoptive NK, CAR-NK, memory-like NK

28 papers · 1996–2025 · 26 with culture/manufacturing content

Clinical trials

- **2005** Miller JS et al. — Successful adoptive transfer and in vivo expansion of human haploidentical NK cells in patients with cancer. *Blood*. PMID 15632206
- **2008** Arai S et al. — Infusion of the allogeneic cell line NK-92 in patients with advanced renal cell cancer or melanoma: a phase I trial. *Cytotherapy*. PMID 18836917
- **2011** Geller MA et al. — A phase II study of allogeneic natural killer cell therapy to treat patients with recurrent ovarian and breast cancer. *Cytotherapy*. PMID 20849361
- **2011** Curti A et al. — Successful transfer of alloreactive haploidentical KIR ligand-mismatched natural killer cells after infusion in elderly high risk acute myeloid leukemia patients. *Blood*. PMID 21791425
- **2014** Bachanova V et al. — Clearance of acute myeloid leukemia by haploidentical natural killer cells is improved using IL-2 diphtheria toxin fusion protein. *Blood*. PMID 24719405
- **2015** Conlon KC et al. — Redistribution, hyperproliferation, activation of natural killer cells and CD8 T cells, and cytokine production during first-in-human clinical trial of recombinant human interleukin-15 in patients with cancer. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*. PMID 25403209
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